The Impact of Wildfire Evacuation on Rental Prices: Evidence from Canada

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Abstract

When wildfires displace households, nearby cities absorb evacuees, and rents may follow. I study this using the July 2017 Williams Lake evacuation in British Columbia, exploiting a panel of 131 Canadian cities from 2008 to 2024. The identification uses a Bartik-style shift-share exposure design that weights cities by their pre-fire migration ties to Williams Lake, with log average two-bedroom rent as the outcome. The baseline reduced-form estimate is 0.818 log-points, implying a 19-percent rent increase in the most exposed destination. City-specific trends reduce it to 0.594. The most demanding test, comparing cities within the same province and year, collapses the estimate to 0.164, indistinguishable from zero, and post-fire migration flows show no amplification toward high-exposure cities. Pre-existing network connections predict rent trajectories, but provincial dynamics confound the wildfire-specific reading. Disaster-relief policy that ignores receiving cities likely underestimates the full housing-cost burden of climate displacement.

Keywords: Wildfires, Climate-induced displacement, Rental prices, Migration networks, Spatial spillovers

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1 Introduction

In July 2017, a wildfire forced roughly 11,000 Williams Lake residents to evacuate with less than 24 hours of warning. Most ended up in Kamloops, Prince George, and Vancouver, drawn by pre-existing family ties and prior residence. Williams Lake was not unusual. The 2023 Canadian wildfire season burned more than 17 million hectares, placing over 230,000 residents under evacuation orders across six provinces (Canadian Interagency Forest Fire Centre, 2024; Public Safety Canada, 2024). Climate projections suggest seasons like 2023 will become routine. Displaced households do not vanish. They move somewhere. Wherever enough of them concentrate, they compete for apartments, and rents rise.

The policy response to wildfire displacement currently stops at the burn perimeter. Emergency management tracks evacuees. Disaster relief flows to affected municipalities. Reconstruction funds target the origin. Receiving cities, where displaced families actually compete for housing, sit outside this framework entirely. If rental pressure from wildfire evacuees is large enough to affect housing affordability, the current policy boundary is drawn too narrowly.

This paper estimates whether Canadian cities with stronger pre-existing migration ties to Williams Lake experienced larger rent increases after the July 2017 evacuation. I use a Bartik-style exposure index that weights each city by its pre-fire share of Williams Lake out-migrants, interacted with a post-2017 indicator, and estimate the effect on log average two-bedroom rents across a balanced panel of 131 Canadian cities from 2008 to 2024. The baseline reduced-form estimate is $\hat{\pi} = 0.818$ log-points (s.e. 0.216), implying a 19 percent post-fire rent differential for Kamloops, the most exposed destination, or roughly \$160 per month on average over 2017–2024 at 2015 rent levels. The most credible specification adds province-by-year fixed effects and yields 0.164 (s.e. 0.260), indistinguishable from zero. Pre-existing migration networks predict rent patterns, but regional confounds prevent a causal reading.

The baseline estimate, taken at face value, is large. Trojanek and Głuszak (2022) find comparable housing market effects from Ukrainian refugee inflows into Poland, a displacement event roughly 100 times larger in absolute terms (1.5 million refugees versus 11,000 evacuees). If a shock two orders of magnitude smaller produces similar point estimates, either the per-capita rental impact of wildfire evacuees is extraordinarily high, or the estimate reflects something other than the fire. The city-specific trends specification brings the coefficient to $\hat{\pi} = 0.594$ (s.e. 0.175), still statistically precise at the one-percent level, but the pre-trend evidence points toward the second interpretation.

Placebo regressions assigned to fictitious treatment years from 2010 to 2016 yield significant positive coefficients in the baseline. Exposed cities were already growing faster before the fire. Province-by-year fixed effects compare only cities within the same province and year. They collapse the estimate to 0.164 and render it insignificant. The Rambachan–Roth breakdown value is $\bar{M}^* = 0.38$. At that threshold, post-treatment deviations from the pre-trend reach 38 percent of the largest pre-treatment violation, and a zero effect enters the confidence set. The baseline coefficient conflates a Williams Lake network effect with the broader British Columbia provincial rent trajectory of the late 2010s.

Annual bilateral migration flows reinforce this reading. Flows from Williams Lake to the 131 destination cities do not rise after the fire. They fall (-95.8, s.e. 8.9), consistent with a general contraction of out-migration from Williams Lake rather than a directed surge toward high-exposure destinations. A placebo test using Quesnel, a nearby forestry town that did not burn, shows no post-2017 shift. Pre-fire migration networks predict post-fire rent trajectories, but the mechanism turns out not to be the direct displacement shock the design was built to detect.

The disaster-housing literature establishes that natural disasters generate lasting out-migration from affected areas. Boustan et al. (2020) document, using a century of American county data, that major disasters cause persistent population loss at the origin, which provides the empirical foundation for the displacement flow this paper traces downstream. For wildfires specifically, home prices rose 13 percent within 25 miles of the 2018 Camp Fire in the six weeks following ignition (Hennighausen and James, 2024). Severe floods raise rents for lowest-income renters in directly affected neighbourhoods by approximately 5 percent in the year after the event (Brennan et al., 2024). A wider set of studies reaches consistent conclusions for hurricanes, coastal flooding, and earthquakes (Vigdor, 2008; Gallagher and Hartley, 2017; Ortega and Taspinar, 2018). Every study in this literature operates inside the damage zone, where supply destruction, insurance payouts, and neighbourhood composition shifts are inextricably confounded with demand pressure. None follows displaced households to the cities where they actually compete for apartments.

A smaller literature examines receiving markets. Harwood (2026) finds heterogeneous rent responses across income groups in United States cities absorbing disaster-displaced households. Lower-income neighbourhoods experience persistent rent declines attributed to renovation financing constraints, while higher-income areas recover more quickly. Crossborder evidence comes from two large displacement events. Trojanek and Głuszak (2022) document short-run housing-market effects of Ukrainian refugee inflows in Poland, and Czerniak (2024) find suggestive rent pressure across European rental markets hosting the same flows. Neither study applies a shift-share design, and both concern international population movements governed by asylum policy rather than voluntary domestic relocation through pre-existing migration networks.

The theoretical prediction for domestic displacement is standard. In a spatial equilibrium, a positive demand shock raises rents in proportion to local supply inelasticity (Saiz, 2010). The empirical benchmark is Saiz (2007), who estimates that an inflow equal to 1 percent of city population raises rents by approximately 1 percent in American metropolitan areas. González and Ortega (2013) document comparable effects on house prices in Spain. British Columbia has mountainous terrain, tight zoning, and short-run supply inelasticity, so the mechanism predicts non-trivial price effects even from moderate inflows. This channel differs from climate gentrification, in which long-run risk-perception shifts revalue lower-elevation or climate-resilient urban land over decades (Keenan et al., 2018; Bernstein et al., 2019). Climate gentrification predicts broad area-wide revaluations independent of network proximity to a specific event. The displacement channel studied here predicts rent increases concentrated in cities with strong pre-fire migration ties to the origin.

Three things distinguish this paper from this literature. First, by estimating rent responses hundreds of kilometres from the burn perimeter, the design isolates a pure demand channel free from supply destruction, risk repricing, and insurance-driven capital flows that confound within-zone estimates. Second, this is the first application of the shift-share framework of Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2022) to domestic climate displacement, linking pre-determined bilateral migration networks to receiving-city rent outcomes. Third, this paper documents precisely where a single-origin design falls short. With exposure concentrated in one small city, the shift-share shares are not plausibly uncorrelated across destinations, and the strongest available test compares only cities within the same province, reducing the estimate from 0.818 to 0.164 without resolving which end of that interval is correct.

Section 2 describes the CMHC rental data, the Statistics Canada bilateral migration flows, and the Williams Lake evacuation context. Section 3 formalises the exposure index and the identifying assumptions under Goldsmith-Pinkham et al.'s framework. Section 4 tests the displacement mechanism directly using annual origin-destination flows, the test that most directly challenges the baseline interpretation. Section 5 presents the main rent estimates and the event-study profile. Section 6 then works through pre-trend placebos, province-year and region-year fixed effects, Rotemberg weight diagnostics, COVID truncation, the Sudbury placebo origin, and the Rambachan–Roth sensitivity bounds. Section 7 concludes.

2 Data and Context

2.1 Williams Lake Evacuation

On July 7, 2017, authorities in Williams Lake, British Columbia issued a mandatory evacuation order with fewer than 24 hours' notice, as a converging firestorm threatened the city during an exceptionally severe wildfire season. Roughly 11,000 residents were ordered to leave. The timing was driven by meteorological and fire-propagation conditions, not by rental-market developments in potential receiving cities, which rules out anticipatory price adjustment by landlords in destination markets. Because the order came without warning, displaced households had no time to search systematically for preferred destinations. They moved toward cities where they already had family ties, prior employment, or a history of residence. The empirical design exploits this by measuring pre-fire migration links between Williams Lake and each Canadian city in the panel.

2.2 Sources and Sample

The panel draws on three Canadian administrative sources. Migration flows come from Statistics Canada Table 17-10-0141-01 (Statistics Canada, 2024), which records annual intercity flows across Census Metropolitan Areas, Census Agglomerations, and related geographies. These bilateral flows construct the pre-fire Williams Lake destination shares and allow a direct test of whether post-fire arrivals from Williams Lake amplified toward high-exposure destinations. Rental data come from CMHC Table 34-10-0133-01 (Canada Mortgage and Housing Corporation, 2024), which reports average monthly rents by market, unit type, and structure type.[1] The CMHC Rental Market Survey for 2017 provides the

[1] All three data sources are publicly available at no cost under the Statistics Canada Open Licence. The complete replication package, including all code and auxiliary files needed to reproduce every table and figure, is available at https://github.com/Robin-Economist/The-Impact-of-Wildfires-on-Rental-Prices-Evidence-from-Canada.

two-bedroom vacancy rates and rental-universe counts used in the economic calibration in Section 5.4. Centroid coordinates for the spatial exposure map are computed from the 2021 Census geography boundary files (Statistics Canada catalogue 92-160-X), accessed via the R package `cancensus` (von Bergmann et al., 2021). Bay Roberts, Leamington, and Cold Lake are geocoded manually from Statistics Canada's 2021 Census community profiles, while the biprovincial agglomerations of Campbellton and Hawkesbury are reconstructed from their provincial geography components to match the migration-table units.

Building the estimation sample requires reconciling two geographic systems that do not map one-to-one. Starting from the 173 areas in the migration table, I drop eight biprovincial subgeographies to avoid double-counting. On the rent side, CMHC markets that correspond to census subdivisions outside any Census Metropolitan Area or Census Agglomeration have no counterpart in the migration table and are excluded. Provincial rural residuals are likewise dropped. The two sources intersect on 145 cities. Thirteen require non-trivial name harmonisation before merging, and four biprovincial cities require province-specific adjustments.

The panel runs from 2008 to 2024. Within the 145-city intersecting sample, the share of city-year observations with non-missing rent averages 98.0% from 2008 onward and never falls below 96.5% in any single year. Coverage falls sharply before 2008, so extending the window back further would trade a longer pre-period for a smaller and less stable city sample.

2.3 Rent Measure

The dependent variable is the logarithm of average monthly rent for two-bedroom apartments in structures containing three or more units (CMHC Table 34-10-0133-01). Following Saiz (2007), who uses a two-bedroom rental measure as the benchmark for primary rental-market conditions in American metropolitan areas, this segment offers the best compromise between economic relevance and coverage in the Canadian data. Two-bedroom rents are non-missing for 98.0% of city-year observations from 2008 onward, while bachelor and three-bedroom series show materially more attrition, particularly in smaller cities. Two-bedroom units also provide a more stable composition proxy than the smallest unit categories, whose mix shifts as developers respond to land costs. Alternative unit and structure definitions appear in Appendix Figure 9.

One interpretive caveat applies throughout. Because CMHC average rent pools longterm controlled leases with new-tenancy rents, and because British Columbia rent regulations cap annual increases for continuing tenants, estimates from this series likely understate the market-clearing response to a demand shock. Wildfire-driven demand shocks manifest first in new leases, which the CMHC average only partially reflects.

2.4 Migration Exposure

The explanatory variable is the pre-fire bilateral migration share $s_{WL,d}$, defined as the fraction of Williams Lake out-migrants who settled in destination city d during the 2016/2017 migration wave:

$$s_{WL,d} = \frac{\text{Migrants from Williams Lake to city } d \text{ in 2016/2017}}{\text{Total out-migrants from Williams Lake in 2016/2017}}. \tag{1}$$

Statistics Canada's migration wave runs from July 2016 to June 2017, placing it entirely before the July 2017 wildfire. Rural residual destinations are excluded and shares are renormalised to sum to one across the retained city destinations. Using the wave immediately preceding the evacuation rather than a multi-year average maximises contemporaneity with the shock. The concentration of Williams Lake out-migrants in a stable set of British Columbia cities suggests that alternative window definitions would yield similar exposure profiles.

Of the 131 cities in the main panel, 43 have a strictly positive pre-fire share and 88 have zero observed share. A caveat: Statistics Canada suppresses bilateral cells below its confidentiality threshold, so genuine small flows toward minor destinations may be recorded as zero. If suppressed cells concentrate among smaller non-British Columbia cities, the zero-exposure control group is contaminated by cities with genuine but unobserved positive exposure, attenuating the estimated differential. The distribution of positive shares is highly concentrated. Kamloops accounts for 21.2%, followed by Vancouver at 17.4%, Prince George at 9.97%, Kelowna at 6.91%, and Quesnel at 6.75%. Across all 131 cities, the mean share is 0.007 and the median is zero.

Two panel consistency adjustments are straightforward. Williams Lake itself enters with $s_{WL,d} = 0$, since out-migrants from a city are not counted as returning to it. For the two biprovincial agglomerations, Campbellton and Hawkesbury, CMHC reports rent by provincial part while the migration data treat each city as a single unit. I average rent across provincial parts and set the migration share to zero, in line with the exclusion of biprovincial sub-geographies on the migration side.

2.5 Policy Context

Three British Columbia housing-policy changes overlap with the post-2017 estimation window and may confound the exposure estimates, particularly for Vancouver and other highly exposed cities. The Foreign Buyers Tax (August 2016) imposed a 15 percent surcharge on residential purchases by foreign nationals in Metro Vancouver, reducing ownership demand and diverting it partly toward the rental market. The Speculation and Vacancy Tax (announced 2018, operative from 2019) extended a similar logic province-wide, adding rental pressure in secondary markets. The Residential Tenancy Act cap on annual rent increases, applied throughout the period but tightened in 2023 to a maximum of 2 percent, limits the observable rent response in tenancies with continuing tenants while leaving new-tenancy rents unrestricted. These policy changes affect the CMHC average-rent series differentially across cities: markets with rapid turnover (Vancouver, Kelowna) display faster adjustment than stable markets with low vacancy. Because the most exposed cities are disproportionately located in British Columbia and subject to these policies simultaneously with the wildfire exposure, the identification threat they pose cannot be eliminated by the baseline design. The Vancouver robustness check in Section 6.5 directly addresses the Vancouver-specific policy channel. The province-by-year fixed effects in Section 6.3 address the broader provincial policy component.

2.6 Sample and Descriptives

The main estimation sample retains 131 cities over 2008–2024. Of the 145-city intersecting sample, 14 cities have at least one missing observation in the baseline rent series and are dropped to preserve a balanced panel. The most exposed among them, Belleville and Courtenay, each account for 0.64% of Williams Lake's pre-fire out-migrants, so their exclusion is unlikely to affect any of the estimates. This is confirmed in the unbalanced-panel exercise in Appendix Table 16. Leamington is the only city with late-period gaps (its rent series drops out in 2023 and 2024) and remains in the sample since these gaps fall well after the core treatment window. The panel is nearly balanced: of $131 \times 17 = 2{,}227$ city-year slots, 2,225 carry non-missing rent observations.

Average monthly rent is $908, with a median of $857 (Table 1). The within-city standard

Table 1: Descriptive Statistics

Panel A: Summary statistics

	Obs.	Mean	Median	Std. Dev.	Min	Max
Monthly rent ($)	2,225	908	857	297	384	2,360
Log monthly rent	2,225	6.76	6.75	0.309	5.95	7.77
$s_{WL,d}$ (%, ×100)	2,225	0.70	0.00	2.70	0.00	21.2

Panel B: Average monthly rent ($) by exposure status and period

	Non-exposed		Exposed	
	Pre-2017	Post-2017	Pre-2017	Post-2017
Mean rent ($)	733	961	895	1,219
Std. dev.	197	284	240	281
Δ ($)	228 (31%)		324 (36%)	
Cities	88		43	
Observations	792	704	387	342

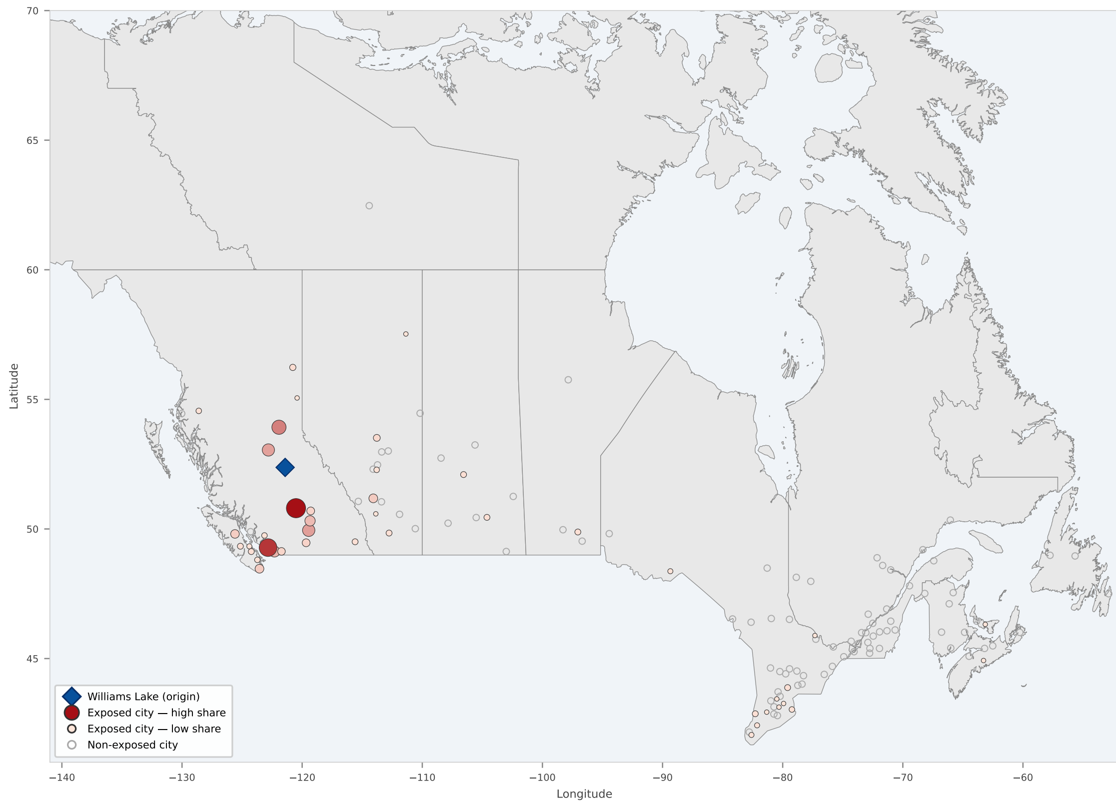
Notes: Unit of observation: city-year. Sample: 131 Canadian cities, 2008–2024 (2,225 observations). Monthly rent is the CMHC average rent for two-bedroom apartments in buildings with three or more units (Table 34-10-0133-01). $s_{WL,d}$ is the pre-fire share of Williams Lake out-migrants settling in city d, from Statistics Canada bilateral flows, wave 2016/2017 (Table 17-10-0141-01); reported in percentage points (×100). Pre-2017: 2008–2016; post-2017: 2017–2024. Exposed cities: $s_{WL,d} > 0$ ($N = 43$); non-exposed: $s_{WL,d} = 0$ ($N = 88$). Panel B reports raw means by exposure status and period.

deviation is $190, against a between-city standard deviation of $229, indicating substantial time variation within markets. Exposed cities ($s_{WL,d} > 0$, $N = 43$) have higher baseline rents than non-exposed cities ($N = 88$) and faster post-2017 growth: mean rents rose from $895 to $1,219 (36%) among exposed cities, against 31% for non-exposed cities. These level differences motivate city fixed effects in all specifications. The differential post-2017 pattern is evaluated formally in Sections 3 and 5.

2.7 Descriptive Evidence

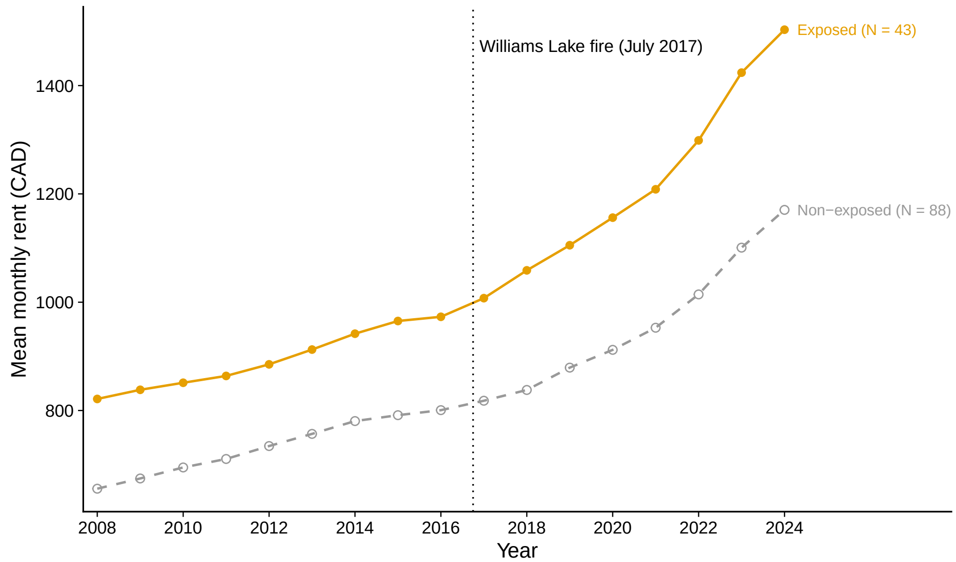
Exposure is geographically concentrated in British Columbia: most Canadian cities have no observed pre-fire link to Williams Lake, while the highest shares cluster in British Columbia (Figure 1). Exposed cities show higher average rents before 2017 and larger post-2017 rent growth than non-exposed cities (Figure 2). Appendix Figures 6 and 7 report the full rent distribution and the histogram of positive migration shares.

Figure 1: Spatial Distribution of Pre-Fire Migration Exposure



Notes: Each point is a city in the main estimation sample ($N = 131$). Filled circles denote the 43 exposed cities ($s_{WL,d} > 0$); colour and size are proportional to the pre-fire migration share $s_{WL,d}$ (share of 2016/2017 Williams Lake out-migrants received by each city). Hollow grey circles are the 88 non-exposed cities ($s_{WL,d} = 0$). The diamond marks Williams Lake, British Columbia (origin). Pre-fire migration shares are from the Statistics Canada 2016/2017 internal migration wave.

Figure 2: Mean Monthly Rent by Exposure Status, 2008–2024



Notes: Annual cross-sectional mean of average monthly rent (CAD) for two-bedroom apartments in structures of three or more units, computed separately for exposed cities ($s_{WL,d} > 0$, $N = 43$, solid line) and non-exposed cities ($s_{WL,d} = 0$, $N = 88$, dashed line). The vertical dotted line marks 2017, the year of the Williams Lake wildfire.

3 Empirical Strategy

Two sources of variation drive the estimates. Across cities, some destinations received a larger share of Williams Lake's pre-fire out-migrants than others, for reasons tied to distance and labour-market proximity rather than to rental conditions at the destination. Over time, the July 2017 evacuation activated those pre-existing links. Cities with stronger historical ties to Williams Lake faced a larger housing-demand shift from 2017 onward. The coefficient $\hat{\pi}$ measures whether that differential exposure translated into faster rent growth after 2017.

3.1 Endogeneity

Estimating how wildfire-displaced households affect rents is harder than it looks. Displaced households do not choose their destination at random: they move toward cities where they have family ties, where labour markets are strong, or where rents are already rising. This selective sorting creates a correlation between observed migrant inflows and unmeasured characteristics of receiving cities, so ordinary least squares estimates of the displacement effect would be unreliable.

Selective sorting biases estimates in both directions. If displaced households move toward dynamic cities whose rental prices are rising for reasons unrelated to the wildfire, ordinary least squares will overstate the effect of displacement. If, instead, they move toward cities perceived as relatively affordable, the same correlation may bias estimates downward. In either case, the observed relation between realised migration flows and rents does not by itself support a causal interpretation.

Including city and year fixed effects partially mitigates this problem. City fixed effects γ_d absorb time-invariant differences in local housing markets, geography, and amenities. Year fixed effects λ_t absorb common macroeconomic shocks affecting all Canadian cities. A two-way fixed-effects regression on observed inflows would take the form

$$\log R_{dt} = \alpha + \beta M_{dt} + \gamma_d + \lambda_t + \varepsilon_{dt},$$

where R_{dt} denotes the average monthly rent in city d and year t, and M_{dt} denotes realised arrivals of displaced households. Even in this framework, M_{dt} may remain correlated with ε_{dt} if destination-specific shocks both attract migrants and raise rents. The empirical strategy therefore replaces realised migration with a predetermined exposure index based on pre-fire destination shares.

3.2 Wildfire Shock

The institutional features of the Williams Lake evacuation reviewed in Section 2.1 provide an aggregate event whose timing is plausibly exogenous to rental-market conditions in potential receiving cities. In the absence of the fire, migration flows between Williams Lake and the rest of Canada would have followed their ordinary path, shaped by labour markets and social networks. The wildfire created an abrupt evacuation shock that may have been transmitted more strongly to cities with closer historical links to the origin. The design exploits this cross-city variation in predetermined exposure rather than realised post-fire mobility itself.

The pre-fire migration shares $s_{WL,d}$ are not distributed randomly across Canadian cities. They reflect the economic geography of the British Columbia Interior. Williams Lake, Quesnel, and Prince George lie along Highway 97, the main north-south corridor of the province's forest economy, at intervals of roughly 120–150 kilometres. Kamloops, 180 kilometres south of Williams Lake via the same highway, is the largest regional service hub for the southern Cariboo. West Fraser Timber and Canfor operate mills across this corridor, creating overlapping labour markets: a mill worker from Williams Lake could find equivalent employment in Quesnel or Prince George without changing industry. When the evacuation order came, residents went to Kamloops, Prince George, or Quesnel because they had worked in those cities before, had family in the same industry there, or had simply driven through on prior occasions. The shares measure this pre-existing economic geography. A gravity model fitted directly to Williams Lake as origin, using destination population and bilateral distance, explains 69 percent of the cross-city variation in observed pre-fire shares ($R^2 = 0.69$, Appendix Figure 13). Replacing observed shares with these gravity-predicted counterparts yields similar main estimates (Table 8, columns 2–3), confirming that $s_{WL,d}$ is driven by observable economic fundamentals rather than idiosyncratic migration choices correlated with destination rental markets.

3.3 Exposure Index

The empirical design uses a Bartik-style exposure index rather than a canonical multi-shock Bartik instrument. In the standard shift-share setting formalised by Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2022), identification can be organised around many independent shocks whose weighted sum varies across locations. Here, the aggregate event is a single wildfire evacuation. Cross-city variation comes entirely from one predetermined exposure profile: the pre-fire destination shares of Williams Lake out-migrants. The relevant identifying assumption is therefore not the exogeneity of a collection of shocks, but the conditional exogeneity of these pre-fire shares, in the sense of Goldsmith-Pinkham et al. (2020), with respect to destination-specific counterfactual rent trajectories.

The migration share $s_{WL,d}$ is defined in equation (1). The exposure variable interacts this predetermined cross-city share with a binary post-treatment indicator:

$$Z_{WL,dt} = s_{WL,d} \times \mathbf{1}[t \geq 2017]. \tag{2}$$

Before the wildfire, $Z_{WL,dt} = 0$ for every city. From 2017 onward, a city's exposure equals its pre-fire migration share $s_{WL,d}$. The 88 cities with $s_{WL,d} = 0$ serve as the low-exposure reference group in this reduced-form design. They had no observed Williams

Lake destination share in the pre-fire wave, though the annual migration data do not justify treating them as receiving literally zero displaced households after the evacuation. This cross-city heterogeneity in exposure intensity, combined with temporal activation at the moment of the shock, supplies the variation used in the main specification. Because the wildfire is a single event shared by all destinations, the shift component $\mathbf{1}[t \geq 2017]$ is constant across cities. All cross-city variation in $Z_{WL,dt}$ therefore comes from the share component $s_{WL,d}$ alone.

The economic interpretation of the exposure index rests on a proportional amplification idea. Cities that historically received a larger share of Williams Lake out-migrants would face proportionally higher exposure to any wildfire-induced relocation shock. Section 4 tests this reading directly against the annual bilateral migration data. High-share destinations do not record the post-fire surge in annual arrivals that a direct displacement interpretation would predict. This negative result does not invalidate the exposure index. Emergency evacuations unfold over days or weeks, while the administrative migration series aggregates flows over twelve-month windows. The displacement signal is too brief and too diffuse to register cleanly in annual records. The exposure index is therefore best understood not as a proxy for an observable flow that happens to be poorly measured, but as a predetermined indirect measure of a process that leaves no clean administrative trace at the available data frequency.

3.4 Specification

The main reduced-form specification regresses log rent on the exposure index $Z_{WL,dt}$ in equation (2), with city and year fixed effects:[2]

$$\log R_{dt} = \alpha + \pi Z_{WL,dt} + \gamma_d + \lambda_t + \varepsilon_{dt}, \tag{3}$$

where γ_d denotes city fixed effects and λ_t denotes year fixed effects. Standard errors are

[2] The decomposition concerns of Goodman-Bacon (2021) and the estimator proposed by Callaway and Sant'Anna (2021) address staggered binary treatment adoption. This design involves a single simultaneous shock: all cities transition from pre-fire to post-fire exposure at the same calendar date (2017). Those critiques do not apply here. A related concern for *continuous* treatments is that the two-way fixed effects estimator may assign negative weights to some average treatment effects when those effects are heterogeneous across intensities (de Chaisemartin and D'Haultfœuille, 2020). For $Z_{WL,dt} = s_{WL,d} \times \mathbf{1}[t \geq 2017]$, the Frisch-Waugh-Lovell decomposition yields TWFE weights proportional to $(s_{WL,d} - \bar{s})^2 \times (\text{Post}_t - T^{\text{post}}/T)^2 \geq 0$ for all (d, t). The simultaneous shock and time-invariant post-treatment intensity guarantee non-negative weights by construction; see Appendix 7.

clustered at the city level to account for serial correlation within destinations.[3] Appendix Table 17 verifies that two-way clustering at the city and province levels does not materially alter inference. The coefficient π measures the post-2017 log-rent differential associated with a one-unit increase in the exposure index. Because no city has $s_{WL,d} = 1$, the economically meaningful objects are effects evaluated over the support of the observed exposure distribution, such as a ten-percentage-point increase in exposure or the share observed for Kamloops.

The error term ε_{dt} captures city-specific shocks to rental demand and supply that are not removed by city or year fixed effects. Several forces belong there. British Columbia housing policies introduced during the sample period (the Foreign Buyers Tax on Metro Vancouver purchases in August 2016, the Speculation and Vacancy Tax announced in 2018) diverted real-estate demand toward the rental market in ways that affected exposed and unexposed cities differently. Commodity price cycles, particularly the oil boom of 2012–2014 and the subsequent bust, generated income shocks for resource-sector workers across the Interior that are not fully absorbed by common year effects. The COVID-19 pandemic accelerated urban-to-secondary-city migration after 2020, disproportionately toward the British Columbia cities that also happen to be most exposed to Williams Lake migration links. Each of these forces is a plausible source of differential rent growth between high-exposure and low-exposure cities, independent of the 2017 wildfire. The identification threat they pose is real and is confronted directly in Section 3.5.

The reduced form is the appropriate estimand given the observability structure of the data. Because emergency displacement is not recorded at the frequency required for a credible first stage, treating the exposure index as an instrument for observed annual inflows would impose a mapping between the predetermined network measure and an endogenous regressor that the data cannot validate. Appendix Section 7 presents an exploratory fixed-effects 2SLS exercise. The first stage is precisely estimated but negative, reflecting mean reversion in annual flows rather than displacement amplification. The reduced form avoids this mismatch by asking whether cities with higher predetermined exposure exhibit different post-2017 rent paths, without requiring that the displacement channel be directly observed at annual frequency. I therefore do not interpret π as a structural elasticity of rents with

[3] Adão, Kolesár, and Morales (2019) show that city-clustered standard errors can be understated in multi-sector Bartik designs when shares are correlated across destinations through common national trends. That concern is structurally limited here: the shares $s_{WL,d}$ reflect the migration geography of a single origin city, not exposure to national sectoral shocks. Two-way clustering by city and province in Appendix Table 17 covers the residual case in which provincial shocks induce cross-city correlation.

respect to realised displaced arrivals.

A second specification adds city-specific linear trends to absorb pre-existing rent trajectories:

$$\log R_{dt} = \alpha + \pi Z_{WL,dt} + \gamma_d + \lambda_t + \delta_d \cdot t + \varepsilon_{dt}. \tag{4}$$

This specification allows each city to follow its own linear rent trajectory over the full sample. The exposure profile is heavily concentrated in British Columbia, where rents were already growing faster than the Canadian average before the wildfire. The city-trend specification partially absorbs these pre-existing trajectories, but it does so by requiring city-level counterfactual rents to be well approximated by linear trends. The estimates with and without city trends are therefore reported as sensitivity benchmarks, not as endpoints of a formal causal interval.

This is a reduced-form exposure specification. The coefficient π is estimated by least squares, but the object of interest is not an OLS regression of rents on realised migration inflows. It is the post-2017 rent differential associated with predetermined Williams Lake exposure, after absorbing permanent city differences and common year shocks.

3.5 Identification

The Bartik-style exposure design rests on three distinct conditions: (i) the wildfire itself must be exogenous to rental-market conditions in receiving cities, (ii) the pre-fire migration shares $s_{WL,d}$ must be orthogonal to counterfactual rent trajectories conditional on the absorbed fixed effects, and (iii) the wildfire must activate the pre-existing exposure profile in a way that is economically relevant for destination rental markets. The first condition is supported by the meteorological nature and emergency timing of the event. The second and third conditions are substantive identifying assumptions whose plausibility is evaluated empirically.

Exogeneity of the aggregate wildfire shock does not by itself make the cross-city exposure profile valid. The deeper problem is that the shares $s_{WL,d}$ are high in British Columbia cities, which were already experiencing faster rent growth than the Canadian average well before 2017. Asking whether cities with stronger Williams Lake ties saw higher rent growth after 2017 is, to a significant degree, asking whether British Columbia cities saw higher rent growth after 2017. They did, and the wildfire is not why. Foreign investment pressure, zoning constraints, a booming tech sector in Vancouver, and net in-migration from other provinces all drove British Columbia rents upward during the same period. The pre-fire migration shares happen to be correlated with this provincial trend, not because Williams Lake evacuees caused it, but because the same economic geography that made certain cities attractive destinations for Williams Lake migrants also made them attractive to everyone else. The placebo tests in Section 6 confirm this reading. Re-estimating the baseline with fictitious treatment years $\tau \in \{2010, \ldots, 2016\}$ produces positive coefficients in most years, meaning the data reject unconditional parallel trends. This is not a routine diagnostic that the design comfortably passes. It is direct evidence that the shares predict rent growth before the fire.

City-specific linear trends in equation (4) absorb part of these pre-existing trajectories, but only under the assumption that counterfactual rents would have continued along a linear path. Province-year fixed effects, which compare only cities within the same province and year, are more demanding. They eliminate the British Columbia-versus-Canada confound directly, at the cost of absorbing any within-province variation that genuinely reflects wildfire exposure. The interpretation of the baseline estimate therefore turns on how much of the post-2017 differential survives once regional rental-market dynamics are absorbed.

The timing of the annual rent series creates a second, more mechanical measurement issue. The wildfire occurred in July 2017, while the Canada Mortgage and Housing Corporation rent series is observed at the annual frequency. Coding $\mathbf{1}[t \geq 2017]$ as the post-treatment indicator assigns the 2017 observation to the exposed period even though rental adjustment may be partial in the first calendar year. The event-study coefficient at $k = 0$ is small and not significant at conventional levels, which is consistent with this timing issue and with lease-based frictions in rent adjustment. Section 6 therefore re-estimates the baseline specification using $\mathbf{1}[t \geq 2018]$ as a direct timing check.

The estimand is reduced form by design, not by default. The variable $Z_{WL,dt}$ is not the realised number of displaced households entering city d. It is a predetermined index of exposure built from historical origin-destination shares. Converting π into a structural elasticity would require observing the displacement flow itself at the relevant frequency, which the available data do not permit. The reduced form sidesteps this unobservability constraint by asking whether cities with stronger predetermined migration links to a disaster origin exhibit systematically different rent trajectories after the disaster. The coefficient π measures this post-2017 rent differential directly. Section 4 examines whether annual bilateral migration flows independently corroborate the displacement reading. Where that corroboration is absent, the interpretation rests on the reduced-form pattern itself.

Taken together, these diagnostics define the scope of the research design. The wildfire supplies plausibly exogenous timing, the pre-fire shares supply predetermined cross-city exposure, and the coefficient π measures a reduced-form rent differential along that exposure profile. The design does not claim to recover a structural elasticity of rents with respect to realised displaced arrivals.

4 Migration Evidence

The rent estimates use a predetermined exposure profile, not observed post-fire migration. The available panel of annual city-to-city flows therefore provides a diagnostic check, not a formal instrument for observed inflows. If the displacement channel operates as the design assumes, destinations with larger pre-fire Williams Lake shares should record systematically larger post-fire annual arrivals from Williams Lake. Table 2 formally tests this prediction. The answer is negative. High-share destinations do not display the post-fire amplification predicted by a direct annual-flow reading of the displacement channel.Appendix Figure 8 reports the corresponding dynamic profiles.

This result limits interpretation. Annual migration records may miss short-lived emergency relocations, and the same pre-fire wave used to define shares can make post-period comparisons prone to a statistical pull toward the average. Even so, the flow data do not directly validate the channel that motivates the exposure design. The rent estimates therefore remain reduced form, documenting a post-2017 rental-market pattern aligned with pre-fire Williams Lake links without establishing that realised displaced inflows are the sole mechanism.

5 Rental Effects

The baseline reduced-form estimate is $\hat{\pi} = 0.818$ (s.e. 0.216). For Kamloops, the most exposed destination with $s_{WL,d} = 0.212$, this point estimate implies a post-2017 rent differential of approximately 19 percent relative to an unexposed city. At the 2015 national reference rent of \$848, that translates to roughly \$160 per month on average over the

Table 2: Annual Migration-Flow Diagnostics

Dependent variable: bilateral annual migration arrivals from origin city. OLS.	(1) Williams Lake	(2) Placebo Quesnel
WL migration share × Post-fire	-95.8^{***}	
	(8.9)	
Quesnel share × Post-fire		30.6
		(29.5)
Observations	655	655
R^2	0.97	0.98
Within R^2	0.12	0.02
RMSE	2.84	2.45
Wave FE	✓	✓
Destination FE	✓	✓

Notes: Each column reports a separate OLS regression of annual bilateral migration arrivals from the origin city to each destination city in the balanced rental-market sample. Column (1): origin is Williams Lake; column (2): origin is Quesnel (placebo). Post-fire is an indicator equal to one for migration waves 2017/2018 onward; pre-fire wave: 2016/2017. Standard errors clustered at the destination city level in parentheses. Wave FE: fixed effects for Statistics Canada annual migration waves (July–June periods), distinct from the calendar-year fixed effects used in the rent regressions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

post-2017 period. All estimates in this section are reduced-form, so $\hat{\pi}$ captures the conditional association between pre-fire exposure and post-2017 rent trajectories rather than a structural causal effect. Section 4 showed that annual migration flows do not validate the displacement channel directly, so the coefficient should be read as a network-correlation pattern, not a clean causal estimate.

5.1 Main Estimates

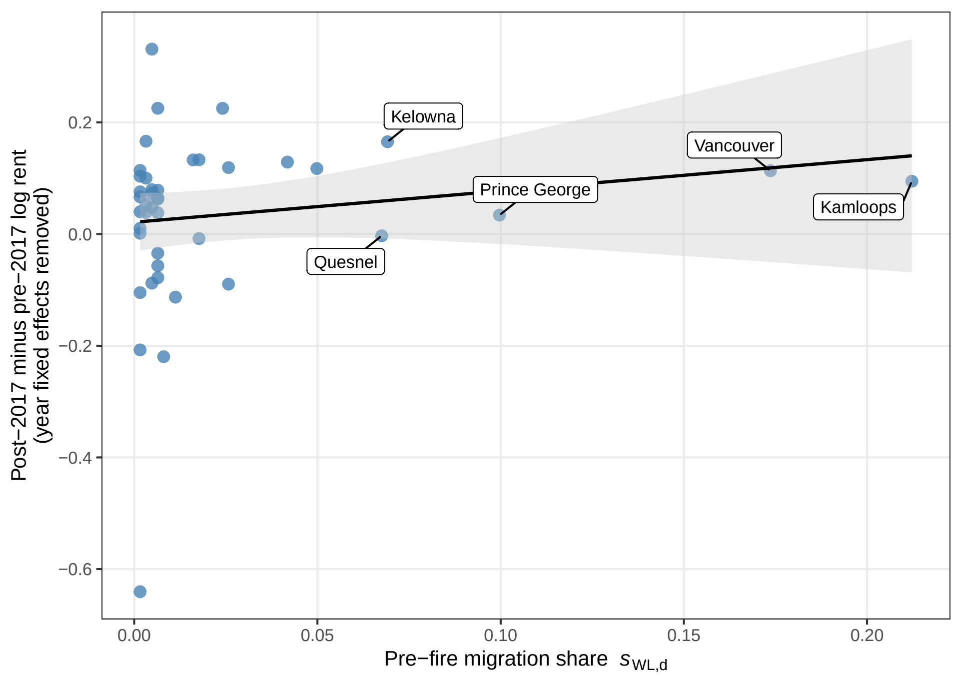
Figure 3 provides the cross-sectional intuition directly. Cities with larger pre-fire Williams Lake migration shares show larger post-2017 rent increases after removing year effects. The slope of that relationship is $\hat{\pi} = 0.818$. Table 3 breaks this down across five specifications.

The raw data already hint at the pattern. Panel B of Table 1 shows that exposed cities saw average monthly rents rise from \$895 to \$1,219 between the pre-2017 and post-2017 periods, a 36 percent increase. Non-exposed cities grew from \$733 to \$961, or 31 percent.

That 5-percentage-point unconditional gap requires no fixed effects. It is simply in the means. The fixed-effects specification tightens this comparison by holding constant permanent city differences and common national trends, and it exploits the continuous variation in exposure shares rather than a binary split. The reduced-form estimate of 0.818 is the residual association once those confounds are absorbed.

The column-by-column pattern in Table 3 shows how much of the raw gap is explained by permanent city differences and common national trends. The no-fixed-effects estimate estimate of 3.84 (1.05) picks up the structural expensiveness of British Columbia cities. Year fixed effects bring it to 2.49. City fixed effects alone yield 3.40. The two-way specification absorbs both, and the coefficient falls to 0.818 (0.216). Adding city-specific linear trends reduces it further to 0.594 (0.175). The two estimates bracket the result from opposite directions.

Figure 3: Reduced-Form Relationship: Pre-Fire Exposure and Post-2017 Rent Changes



Notes: Each point is an exposed city ($s_{WL,d} > 0$, $N = 43$). The x-axis plots the pre-fire Williams Lake migration share $s_{WL,d}$ (in decimal units); the y-axis plots the post-minus-pre change in log monthly rent (log points) after removing year fixed effects. The solid line shows the least-squares fit of the reduced-form exposure relationship. The five most exposed destinations are labelled.

Table 3: Reduced-Form Rent Effects by Williams Lake Exposure

	(1) No FE	(2) Year FE	(3) City FE	(4) City + Year FE	(5) City Trends
WL exposure index	3.838***	2.488***	3.397***	0.818***	0.594***
	(1.050)	(0.856)	(0.848)	(0.216)	(0.175)
Observations	2,225	2,225	2,225	2,225	2,225
R^2	0.06	0.32	0.66	0.94	0.98
Within R^2		0.03	0.07	0.02	0.01
RMSE	0.300	0.254	0.179	0.075	0.049
Year FE		✓		✓	✓
City FE			✓	✓	✓
City trends					✓

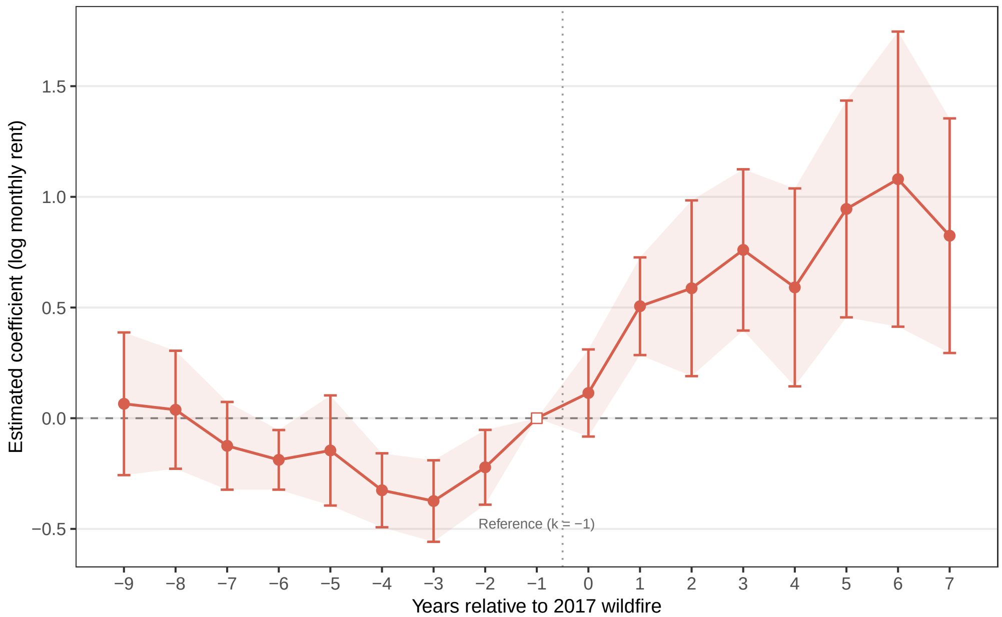
Notes: This table reports reduced-form exposure estimates. The dependent variable is log monthly rent for two-bedroom apartments in buildings with at least three units. The Williams Lake exposure index is the predetermined pre-fire destination share $s_{WL,d}$ interacted with a post-2017 indicator. Column (5) includes city-specific linear time trends. Standard errors are clustered at the city level in parentheses. Period: 2008–2024. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Dynamics

The July 2017 evacuation order reached Williams Lake residents with less than 24 hours of warning, providing a sharp demand shock to the migration networks documented in Section 2. Figure 4 tracks how rents in connected cities evolved year by year relative to 2016. Each coefficient $\hat{\beta}_k$ comes from a fully interacted specification that regresses log rent on $s_{WL,d} \times \mathbf{1}[\text{year} = 2017 + k]$, with city and year fixed effects and the final pre-fire year $k = -1$ omitted as the reference period.

The pre-treatment coefficients for $k \in \{-9, \ldots, -2\}$ are negative in several periods and some confidence intervals exclude zero (the largest deviation in absolute value is $\hat{\beta}_{-3} = -0.374$ at $k = -3$, s.e. 0.094). The pattern shows no monotonic divergence toward higher rents before 2017. The most plausible interpretation is that 2016, the reference year, coincided with a province-wide rent acceleration that makes earlier years appear relatively lower for exposed cities when measured against that baseline. The event study examines year-specific deviations relative to 2016, whereas the placebo-treatment design in Section 6 tests whether the cumulative binary jump structure falsely detects effects in earlier years. These diagnostics target different objects and are both informative. Anticipation effects

Figure 4: Event Study: Williams Lake Exposure and Rental Prices



Notes: Coefficients $\hat{\beta}_k$ from a fully interacted specification that regresses log rent on $s_{WL,d} \times \mathbf{1}[\text{year} = 2017 + k]$, with city and year fixed effects. The omitted reference year is 2016 ($k = -1$). Shaded bands denote 95% confidence intervals. Standard errors are clustered at the city level. Dependent variable: log monthly rent for two-bedroom apartments in buildings with at least three units. Period: 2008–2024.

are implausible in this setting. The evacuation order was issued with less than 24 hours of notice, leaving no window for property owners or tenants in receiving cities to price in the shock in advance.

The coefficient at $k = 0$ is 0.256 (s.e. 0.162), with a 95 percent confidence interval of $[-0.062,\ 0.574]$. This is an imprecise null rather than a precisely estimated zero, as the upper bound of the interval reaches values comparable to those observed at $k = 2$ and beyond. The most likely mechanical explanation is that Canada Mortgage and Housing Corporation measures rents in October, three months after the July wildfire, so lease-based adjustment would not be complete within a single calendar year. From $k = 1$ onward, coefficients grow and remain positive, with $\hat{\beta}_1 = 0.648$ (s.e. 0.150), $\hat{\beta}_3 = 0.902$ (s.e. 0.202), and $\hat{\beta}_6 = 1.222$ (s.e. 0.348; 95% CI $[0.540,\ 1.904]$). For Kamloops, $\hat{\beta}_6$ implies a differential of roughly $1.222 \times 0.212 \approx 0.259$ log-points, or about 29.5 percent in levels, relative to an unexposed city.

This dynamic pattern is compatible with a gradual receiving-market response, but it is also compatible with the regional pre-trend concern identified in the placebo tests. A one-time evacuation lasting approximately three weeks cannot by itself explain rent differentials that persist and grow for seven years. The post-2020 acceleration is partly attributable to COVID-era migration toward British Columbia secondary cities, as confirmed by the truncation check in Section 6.6, which reduces the estimate from 0.818 to 0.544 when the sample is restricted to 2019. The event-study evidence therefore strengthens the descriptive pattern, not the causal attribution.

5.3 Segment Heterogeneity

Displaced households arrive with different family compositions, from single workers to families with children, so demand pressure from the Williams Lake evacuation should not be confined to two-bedroom units. The ex-ante prediction serves as a breadth check. If the two-bedroom baseline reflects genuine market-wide pressure rather than a segment-specific pricing pattern, comparable coefficients should appear across other apartment types. Appendix Figure 9 confirms this. Positive coefficients of comparable magnitude appear across the three well-covered segments. Row-house cells are too sparse to carry weight. This breadth check does not address the paper's central identification concern and is therefore reported in Appendix Figure 9. To address multiple testing concerns, I examine four dimensions of heterogeneity in total, covering apartment segment, city size, East-versus-West region, and housing supply constraints following Saiz (2010). Appendix Figure 10 plots the point estimates and 95% confidence intervals across all seven subgroups, visualizing the full set of cuts regardless of significance. None contradicts the pooled estimate.

5.4 Economic Magnitudes

The reduced-form coefficients translate into city-level magnitudes through the observed exposure shares (Table 4). Under the baseline estimate $\hat{\pi} = 0.818$, the implied log-rent differential, its percentage equivalent, and its dollar equivalent appear in the first set of columns. The final column repeats the calculation using the city-trend coefficient $\hat{\pi} = 0.594$. These values are scale conversions of the reduced-form association, not estimates of a structural rent elasticity.

The same table also reports a deliberately simple potential-demand scenario. It starts from the 11,000 Williams Lake evacuees, allocates them across destinations using the pre-fire exposure shares $s_{WL,d}$, assumes that a fraction $q = 0.25$ generates rental demand in receiving cities, and converts persons into households using $h = 2$ persons per unit. Under this illustrative scenario, the implied number of renter households is approximately 292 in Kamloops, 239 in Vancouver, 137 in Prince George, 95 in Kelowna, and 93 in Quesnel. These numbers are not realised flows and do not contradict the negative annual-flow diagnostic in Section 4. They provide only an order-of-magnitude translation of the exposure profile into potential demand counts.

Table 4: Economic Calibration of the Reduced-Form Magnitudes

City	Share[a]	Displaced[b]	Households[c]	Log-point effect[d]	Δ% baseline[e]	Δ$ baseline[f]	Δ% trend[g]
Kamloops	0.212	2,334	292	0.174	19.0	161	13.4
Vancouver	0.174	1,910	239	0.142	15.3	129	10.9
Prince George	0.100	1,096	137	0.082	8.5	72	6.1
Kelowna	0.069	760	95	0.057	5.8	49	4.2
Quesnel	0.068	743	93	0.055	5.7	48	4.1

Notes: Top-five most exposed cities in the main sample. [a] $s_{WL,d}$: pre-fire Williams Lake migration share. [b] $11{,}000 \times s_{WL,d}$: evacuees allocated to city d under full dispersion. [c] Potential renter households under the illustrative scenario $(q, h) = (0.25,\ 2)$: $\widehat{H}_d = q \times 11{,}000 \times s_{WL,d}/h$. [d] $\hat{\pi} \times s_{WL,d}$ using $\hat{\pi} = 0.818$ from column (4) of Table 3. [e] $(\exp(\hat{\pi}\, s_{WL,d}) - 1) \times 100$. [f] $\Delta\% \times \$848$, the 2015 CMHC national average monthly rent (pre-fire reference, all 131 sample cities). [g] Same calculation using $\hat{\pi} = 0.594$ from column (5) of Table 3. Household counts are scenarios, not estimates of realised post-fire relocations.

Potential-demand counts can be compared with immediately available rental capacity in each destination (Table 5). For each of the five most exposed cities, the table uses the October 2017 Canada Mortgage and Housing Corporation Rental Market Survey count of private two-bedroom apartment units and the corresponding two-bedroom vacancy rate. This segment closely matches this paper's main outcome. Implied vacant units equal the two-bedroom rental universe multiplied by the vacancy rate. The final columns report the ratio of potential renter households to vacant two-bedroom apartment units under three scenarios: $q = 0.10$, $q = 0.25$, and $q = 0.50$, with $h = 2$ in each case.

The ratios are large in small markets. Under the intermediate scenario, potential renter households amount to roughly 18 times the stock of vacant two-bedroom apartments in Kamloops, about 20 times in Kelowna, nearly 10 times in Quesnel, and about 3 times in Prince George. Vancouver is the exception. Its two-bedroom rental stock is large enough that the same scenario implies a ratio below one. These calculations show that the reduced-

Table 5: Potential Rental Demand Relative to Vacant Two-Bedroom Apartments in 2017

City	Units[a]	Vacancy[a]	Implied vacant	Households[b]	Households / vacant units $q = 0.10$	$q = 0.25$	$q = 0.50$
Kamloops	1,499	1.1%	16.5	292	7.1	17.7	35.4
Vancouver	26,375	1.0%	263.8	239	0.4	0.9	1.8
Prince George	1,505	3.0%	45.1	137	1.2	3.0	6.1
Kelowna	2,341	0.2%	4.7	95	8.1	20.3	40.6
Quesnel	318	3.0%	9.5	93	3.9	9.7	19.5

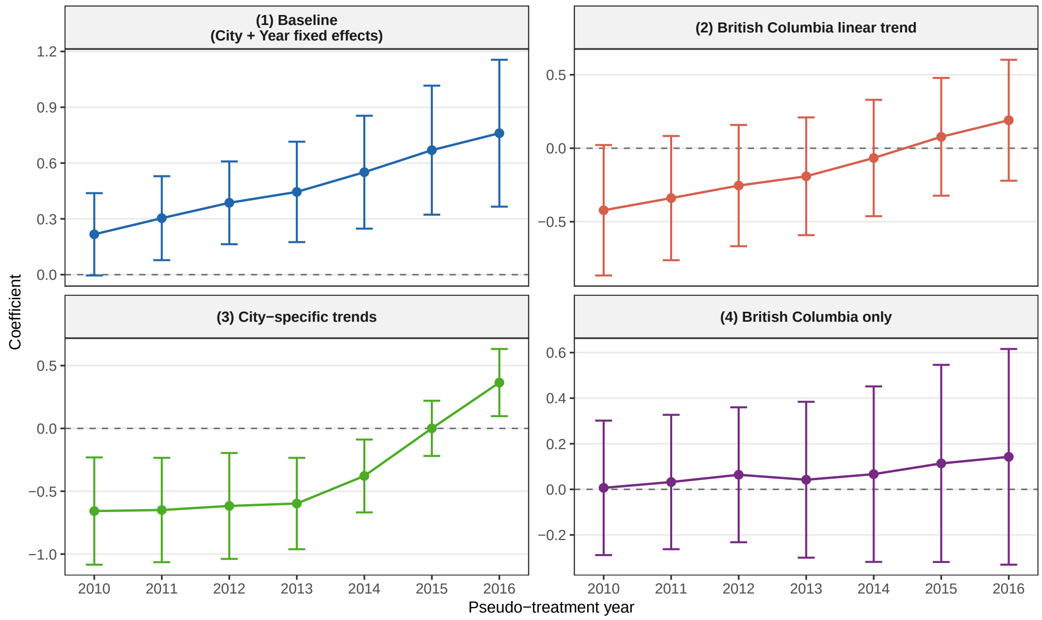
Notes: [a] Two-bedroom units and vacancy rates from the Canada Mortgage and Housing Corporation (CMHC) 2017 British Columbia Rental Market Report, private apartment structures with at least three rental units, measured October 2017. Implied vacant units equal two-bedroom units multiplied by the vacancy rate. [b] Potential renter households under the intermediate scenario $(q, h) = (0.25, 2)$ from Table 4. H/V columns report potential renter households divided by implied vacant two-bedroom units under three scenarios for the share of evacuees generating rental demand (q), with household size $h = 2$. This table is a market-tightness plausibility exercise, not an estimate of realised migration or a test of the identification strategy.

form magnitudes are not mechanically implausible for smaller, highly exposed markets that were already very tight in October 2017. They do not establish that displaced Williams Lake households generated the rent differentials estimated in Section 5. Conversely, the large household-to-vacant-unit ratios, reaching 17.7× for Kamloops under the intermediate scenario, suggest that a pure displacement channel operating through standard market clearing would generate price pressures exceeding the 19 percent baseline estimate.Because the calibration starts from the roughly 11,000 Williams Lake evacuees rather than the broader regional population affected by the 2017 wildfire season, it should be interpreted as a conservative lower-bound scenario for potential receiving-market demand, conditional on the displacement interpretation.

6 Robustness

This section focuses on the three diagnostics that matter most for interpretation, covering pre-trend violations, regional confounding, and the specificity of the Williams Lake exposure profile. Secondary checks are reported in Appendix Tables 17 and 16.

Figure 5: Placebo Treatment-Year Diagnostics



Notes: Each panel plots the coefficient on $\tilde{Z}_{WL,dt}(\tau) = s_{WL,d} \times \mathbf{1}[t \geq \tau]$ for $\tau \in \{2010, \ldots, 2016\}$. The panels correspond to: (i) city and year fixed effects, (ii) the baseline augmented with a British Columbia-specific linear trend, (iii) the baseline augmented with city-specific linear trends, and (iv) the British Columbia subsample. Bars show 95% confidence intervals. Standard errors are clustered at the city level. Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. Period: 2008–2024.

6.1 Placebo Tests

The placebo treatment-year exercise is the central diagnostic for the baseline design. For each fictitious treatment year $\tau \in \{2010, \ldots, 2016\}$, equation (3) is re-estimated after replacing $\mathbf{1}[t \geq 2017]$ with $\mathbf{1}[t \geq \tau]$. If the baseline coefficient were driven only by the post-2017 Williams Lake event, these placebo exposures should not systematically predict rent growth before the actual wildfire.

Figure 5 shows that this implication fails in the baseline specification. All seven placebo coefficients are positive, they become larger as the fictitious treatment year approaches 2017, and most are statistically distinguishable from zero. This pattern indicates that cities with greater Williams Lake exposure were already on different rent trajectories before the wildfire. Given the geographic concentration of exposure, the most plausible reading is a British Columbia trend component correlated with the exposure profile.

Additional controls partially attenuate the pre-trend imbalance but do not erase it. Adding a British Columbia-specific linear trend makes the placebo coefficients imprecise and reduces the main post-2017 coefficient to 0.261 (0.214) in Table 6. City-specific linear trends also weaken the placebo pattern, but they rely on a stronger functional-form assumption for the counterfactual rent path. Restricting the sample to British Columbia does not reveal significant placebo coefficients, but that subsample contains only 24 cities and therefore has limited power. The appropriate conclusion is not that within-province parallel trends are established, but that the strongest evidence of pre-trend imbalance operates across provinces.

The checks do not isolate a preferred estimate. The estimates in Table 6 range from 0.144 to 0.873 across specifications, reflecting genuine uncertainty about the counterfactual trend path.

Table 6: Identification and Specification Checks

	(1) Baseline	(2) City trends	(3) BC trend	(4) Province trends	(5) BC only	(6) Unbalanced	(7) WL+FM	(8) $t \geq 2018$
Z_{WL}	0.818***	0.594***	0.261	0.298	0.144	0.828***	0.835***	0.873***
	(0.216)	(0.175)	(0.214)	(0.219)	(0.260)	(0.218)	(0.234)	(0.222)
Z_{FM}							−0.670	
							(0.813)	
Observations	2,225	2,225	2,225	2,225	408	2,382	2,242	2,225
R^2	0.94	0.98	0.94	0.95	0.92	0.94	0.95	0.94
Within R^2	0.02	0.01	0.00	0.00	0.00	0.02	0.03	0.02
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
City FE	✓	✓	✓	✓	✓	✓	✓	✓
City trend		✓						
BC trend			✓					
Province trend				✓				

Notes: Dependent variable: log monthly rent for two-bedroom apartments in buildings with at least three units. $Z_{WL} = s_{WL,d} \times \text{Post}_{2017}$ is the Williams Lake Bartik exposure index. Column (3) adds a British Columbia-specific linear time trend. Column (4) adds province-specific linear time trends. Column (5) restricts the sample to British Columbia cities ($N = 408$ city-year observations). Column (6) uses the unbalanced panel retaining all cities with at least one non-missing rent observation ($N = 2{,}382$). Column (7) adds Z_{FM}, the Fort McMurray exposure index based on gravity-predicted destination shares interacted with Post_{2016}. Column (8) replaces Post_{2017} with $\mathbf{1}[t \geq 2018]$. All specifications include city and year fixed effects. FE: fixed effects. Standard errors clustered at the city level in parentheses. Within $R^2 = 0.00$ in columns (3)–(4) indicates that Z_{WL} explains no residual within-cell variation once province-specific or British Columbia linear trends are absorbed, consistent with the regional confounding hypothesis. Period: 2008–2024. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Trend Sensitivity

The placebo evidence motivates an explicit sensitivity analysis rather than a binary accept-or-reject treatment of parallel trends. Table 11 in the Appendix implements the honest-inference framework of Rambachan and Roth (2023). The analysis asks how large post-treatment deviations from parallel trends may be, relative to the largest observed pre-treatment deviation, before the post-2017 rent differential can no longer be distinguished from zero.

The breakdown value reported in the Appendix is $\bar{M}^* = 0.38$. In substantive terms, the post-2017 confidence set ceases to exclude zero once future trend violations are allowed to reach roughly two-fifths of the largest pre-treatment deviation. This is not a reassuring robustness margin. It quantifies the same point made visually by the placebo plots. The rent pattern is statistically sharp under the baseline specification, but its causal interpretation is sensitive to modest departures from the identifying trend restriction.

6.3 Regional Confounding

The central identification challenge is that nearly all exposed cities are located in British Columbia, which experienced provincial rent growth systematically above the Canadian average from 2010 onward. The province-by-year fixed effects specification addresses this directly by comparing only cities within the same province and calendar year, absorbing any province-level annual shock, including the combined effect of British Columbia housing policies and broader demand dynamics documented in Section 2.5. Under this specification, the coefficient falls to 0.164 (0.260) (Table 7, column 2). This is the most important single number in the robustness analysis. The 95 percent confidence interval runs from approximately -0.35 to $+0.49$, which spans both zero and effects as large as the baseline estimate. This is not a precisely estimated zero. The province-year specification absorbs most of the identifying variation with only 24 British Columbia cities in the sample, so the test cannot rule out effects of the magnitude documented in columns (1) and (5). What it does establish is that the data no longer detect a Williams Lake exposure differential at conventional significance levels once the provincial rent trajectory is fully absorbed.

This result does not imply that the baseline estimate is spurious. Province-by-year fixed effects are also very demanding. With only 24 British Columbia cities in the sample and annual shocks that are province-wide, they absorb substantial within-province variation that may genuinely reflect the wildfire. The test has limited power. Region-by-year fixed effects, which group British Columbia with Alberta, Saskatchewan, and Manitoba into a single "West" region, are less saturated and preserve a sizeable differential of 0.957 (0.308), suggesting the baseline is not merely a West-versus- non-West contrast. The within-Western-province subsample ($\hat{\pi} = 0.933$, s.e. 0.303) reinforces this. The appropriate reading is that the truth lies somewhere between these bounds. The British Columbia provincial trend explains part of the baseline differential, but not necessarily all of it.

Table 7: Regional Absorption and Exposure Concentration

Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. Reduced-form exposure specification estimated by OLS..

	(1) Baseline	(2) Province×Year	(3) Region×Year	(4) Within-West
Z_{WL} (BC 2017)	0.818***	0.164	0.957***	0.933***
	(0.216)	(0.260)	(0.308)	(0.303)
Observations	2,225	2,191	2,225	867
R^2	0.94	0.96	0.94	0.85
Within R^2	0.02	0.00	0.03	0.03
RMSE	0.075	0.065	0.075	0.103
City FE	✓	✓	✓	✓
Year FE	✓			✓
Province×Year FE		✓		
Region×Year FE			✓	

Notes: Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. $Z_{WL} = s_{WL,d} \times \text{Post}_{2017}$: Williams Lake shift-share exposure index. Column (2): province-by-year fixed effects absorb common year effects; 34 singleton observations dropped (N = 2,191). Column (3): region-by-year fixed effects following Statistics Canada geographic divisions (West = British Columbia, Alberta, Saskatchewan, Manitoba; East = Ontario, Quebec, New Brunswick, Nova Scotia, Prince Edward Island, Newfoundland and Labrador; territories excluded from the CMHC sample). Column (4): Western provinces subsample ($N_{\text{cities}} = 51$). Standard errors clustered at the city level in parentheses. Period: 2008–2024.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 Timing and Specificity

Two targeted checks address narrower alternatives. First, redefining the post indicator as $\mathbf{1}[t \geq 2018]$ yields 0.873 (0.222) in column (8) of Table 6. The baseline result is therefore not created by classifying the partial 2017 observation as post-fire.

Table 8 reports three additional checks on the exposure definition. Columns (2) and (3) replace observed pre-fire migration shares with gravity-predicted shares based on destination population and bilateral distance. Column (2) uses OLS log-linear gravity, yielding 2.670 (s.e. 0.916). Column (3) uses Poisson Pseudo-Maximum Likelihood (Santos Silva and Tenreyro, 2006), which handles zero flows, yielding a coefficient of 1.317 (s.e. 0.522). Both specifications produce statistically significant positive coefficients, confirming that the rent pattern does not depend on the specific migration share measure used.

The Quesnel placebo origin returns 0.682 (0.348) with $p = 0.052$ in Table 8. Because Quesnel is geographically and economically close to Williams Lake, the placebo is not a clean null origin. Precisely for that reason, its non-negligible coefficient weakens a Williams Lake-specific interpretation and points toward a broader British Columbia connectivity component.

Table 8: Timing and Specificity

Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. Reduced-form exposure specification estimated by OLS..

	(1) Baseline	(2) Gravity OLS	(3) Gravity PPML	(4) Placebo Quesnel
Z_{WL} (observed shares)	0.818***			
	(0.216)			
Z_{WL}^{grav} (gravity OLS)		2.670***		
		(0.916)		
$Z_{WL}^{grav,ppml}$ (gravity PPML)			1.320**	
			(0.522)	
$Z_{Quesnel}$ (placebo shares)				0.682*
				(0.348)
Observations	2,225	2,225	2,225	2,225
R^2	0.940	0.941	0.940	0.940
Within R^2	0.021	0.027	0.016	0.013
RMSE	0.075	0.075	0.075	0.076
City FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: Column (1): baseline specification with observed pre-fire migration shares. Column (2): exposure index replaced by gravity-predicted shares $\hat{s}_{WL,d}^{grav} \propto \text{Pop}_d/\text{dist}(WL,d)^{\hat{\beta}_1}$ estimated by OLS log-linear gravity. Column (3): same shares predicted by Poisson Pseudo-Maximum Likelihood (Santos Silva and Tenreyro, 2006), which handles zero flows and corrects heteroskedasticity in multiplicative gravity models. Column (4): placebo exposure index using Quesnel (BC) pre-fire migration shares $\times$ Post_{2017}; Quesnel did not burn in 2017. All specifications include city and year fixed effects. Standard errors clustered at the city level in parentheses. Period: 2008–2024. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.5 The Vancouver Case

Vancouver is both highly exposed and subject to contemporaneous housing-market pressures, so a one-city policy channel deserves direct scrutiny. Excluding Vancouver addresses the concern that post-2017 rent increases in that city may reflect concurrent British Columbia housing policies (Foreign Buyers Tax 2016, Speculation and Vacancy Tax 2018) rather than wildfire exposure. The Vancouver $\times$ Post_{2017} indicator in column (3) absorbs any post-2017 shock specific to that city without restricting its contribution to identification. Dropping Vancouver, or absorbing a Vancouver-by-post-2017 interaction, leaves the Williams Lake coefficient at 0.874 (0.326) in Table 9. Removing both Kamloops and Vancouver raises the coefficient to 1.596 (0.641), which argues against a purely two-city story

Table 9: Vancouver and Policy Confounding

Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. Reduced-form exposure specification estimated by OLS..

	(1) Baseline	(2) Drop Vancouver	(3) Vancouver×Post	(4) Drop Kamloops+Van.
Z_{WL} (BC 2017)	0.818***	0.874***	0.874***	1.600**
	(0.216)	(0.326)	(0.326)	(0.641)
Vancouver × Post_{2017}			−0.032	
			(0.055)	
Observations	2,225	2,208	2,225	2,191
R^2	0.94	0.94	0.94	0.94
Within R^2	0.02	0.02	0.02	0.02
RMSE	0.075	0.076	0.075	0.076
City FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. $Z_{WL} = s_{WL,d} \times \text{Post}_{2017}$: Williams Lake shift-share exposure index; column (1) replicates column (4) of Table 3. Column (2): Vancouver excluded from the estimation sample (N = 2,208). Column (3): adds a Vancouver × Post_{2017} indicator without removing the city. Column (4): Kamloops ($s_{WL} = 0.212$) and Vancouver ($s_{WL} = 0.174$) both excluded, the two highest-exposure cities representing approximately 40% of total identification weight (N = 2,191). All specifications include city and year fixed effects. Standard errors clustered at the city level in parentheses. Period: 2008–2024.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

while also showing that the coefficient is sensitive to support changes in the exposure distribution. This check attenuates single-city policy concerns. It does not resolve province-wide British Columbia confounding.

6.6 Additional Checks

Table 10 reports three diagnostics that target the monotonic post-2017 growth trajectory and the non-specificity of the Quesnel placebo.

The COVID-19 pandemic pushed large-scale out-migration from metropolitan areas toward secondary British Columbia cities, which are precisely the cities most exposed in this sample. Truncating the sample at 2019 isolates the pre-pandemic window. The pre-COVID coefficient in column (2) of Table 10 is 0.544*** (s.e. 0.177), positive, significant, and two-thirds of the baseline. The post-2017 pattern is not a pandemic artefact. The attenuation from 0.818 to 0.544 does suggest that COVID-era migration toward British Columbia secondary cities amplified the full-sample signal.

The Quesnel placebo fails as a specificity test because cities that receive migrants from Quesnel heavily overlap with those that receive migrants from Williams Lake. A cleaner test uses Greater Sudbury (Ontario), a resource-sector city approximately 5,000 kilometres from Williams Lake with an orthogonal migration network. The Sudbury placebo coefficient is 0.548*** (s.e. 0.208), column (4). The pre-COVID Williams Lake estimate (0.544, s.e.

Table 10: Robustness: COVID Truncation, Pre-Period Trends, and Distant Placebo

Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. Reduced-form exposure specification estimated by OLS..

	(1) Baseline	(2) Pre-COVID ($\leq$2019)	(3) Pre-period trends	(4) Placebo Sudbury	(5) Placebo Quesnel
Z_{WL} (BC 2017)	0.818***	0.544***	1.060***		
	(0.216)	(0.177)	(0.309)		
Z_{Sudbury}				0.548***	
				(0.208)	
Z_{Quesnel}					0.682*
					(0.348)
Observations	2,225	1,572	2,225	2,225	2,225
R^2	0.940	0.962	0.897	0.940	0.940
Within R^2	0.021	0.015	0.027	0.005	0.013
RMSE	0.075	0.052	0.086	0.076	0.076
City FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓

Notes: Dependent variable: log monthly rent, two-bedroom apartments, buildings $\geq$ 3 units. Column (1): baseline reduced-form (city and year FE), 2008–2024. Column (2): sample truncated at 2019 to exclude COVID-era mobility shocks toward British Columbia secondary cities. Column (3): log rent detrended using city-specific linear trends estimated on pre-treatment data (2008–2016) only; reduced-form on detrended outcome. Column (4): placebo origin Greater Sudbury, Ontario, approximately 5,000 km from Williams Lake, with an orthogonal migration network; pre-fire 2016/2017 shares $\times$ Post_{2017}. Column (5): placebo origin Quesnel, British Columbia, geographically proximate to Williams Lake; pre-fire 2016/2017 shares $\times$ Post_{2017}. Standard errors clustered at the city level in parentheses. Period: 2008–2024 (columns 1, 3–5) or 2008–2019 (column 2). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

0.177) and this Sudbury coefficient differ by less than one standard error of either. A city with no British Columbia connection and no wildfire generates the same coefficient as Williams Lake under its cleanest pre-pandemic specification. The exposure index does not isolate a mechanism specific to the 2017 fire. This result reinforces the province-by-year fixed effects evidence. The data document a reduced-form rent pattern correlated with pre-existing migration connectivity, but that pattern cannot be attributed to Williams Lake in particular.

The city-trend specification in equation (4) estimates trends over the full 2008–2024 sample. When the treatment has a persistent positive effect, the estimated slope for exposed cities is contaminated upward by that effect, which biases $\hat{\pi}$ downward. Estimating city-specific slopes from pre-treatment data only (2008–2016) and extrapolating forward removes this contamination. The reduced-form estimate on the detrended outcome is 1.06*** (s.e. 0.309), column (3), larger than the baseline 0.818 and substantially larger than the full-sample city-trend estimate of 0.594. The standard city-trend specification absorbs part of the treatment effect into the estimated slope. The pre-period-only estimate is an upper bound relative to the city-trend specification, though it inherits the regional confounding concerns that run across all full-sample designs.

Taken together, the three checks do not resolve the identification ambiguity. The pre-

COVID estimate rules out a purely pandemic-driven reading. The Sudbury equivalence undercuts a wildfire-specific reading. The pre-period detrending places the city-trend estimate on its correct side of the baseline. Rotemberg weights show that Kamloops and Vancouver together account for 44 percent of the identifying variation. Appendix A.7 shows that removing Kamloops raises the estimate to 1.100, and dropping the ten most exposed cities yields 5.420 (s.e. 2.700), arguing against a single-city story.

A sharper comparison comes from Fort McMurray. That 2016 wildfire displaced approximately 90,000 residents, eight times the Williams Lake evacuation, yet the Fort McMurray exposure index yields a coefficient of 0.055 (s.e. 1.000), column (2) of Table 16. The estimate is too imprecise to support inference about the underlying mechanism. Whether this reflects rapid return of evacuees, a different destination network, or insufficient identifying variation in the Fort McMurray exposure index is not determined by these data. The contrast is nonetheless informative. The shift-share design does not mechanically assign positive rent effects to every wildfire-exposed city. Alternative exposure specifications and influence diagnostics appear in Appendix Table 16 and Figures 12, 9, 13–14.

7 Conclusion

This paper documents that cities with stronger pre-fire migration ties to Williams Lake experienced differentially higher rents after the July 2017 wildfire. The baseline estimate implies a 19-percent rent increase in the most exposed destination, falling to 13 percent with city-specific trends. The pattern holds across several demanding specifications. It breaks down when the comparison is restricted to cities within the same province, which bounds the wildfire-specific estimate to the interval [0.164, 0.818] rather than pinning it down.

The distributive dimension of this pattern deserves explicit attention. The cost falls on existing renters in destination cities, households with no connection to the event. A family already renting in Kamloops faces higher housing costs because of a fire 350 kilometres away. Evacuees seeking temporary shelter in destination cities compete with these residents for a tight rental stock. The persistence of the rent differential through 2024 suggests this cost is non-trivial and falls disproportionately on low-income renters in small, supply-constrained markets.

Adaptation policy currently stops at the burn perimeter. Emergency management tracks evacuees. Reconstruction funds flow to origin cities. Receiving cities, where displaced families actually compete for housing, sit outside this framework. The data here are consistent with displacement pressure transmitting through migration networks to rental markets hundreds of kilometres from the fire, a channel that existing policy tools do not reach.

Whether this network channel generalises beyond Canadian wildfire evacuations to floods, heat events, or sudden displacements in thin rental markets elsewhere remains open. The Sudbury result suggests the active mechanism is migration connectivity rather than fire-specificity, which widens the potential scope of the pattern but also raises the bar for isolating disaster-specific effects.

Higher-frequency mobility data would permit sharper identification of the timing between evacuation and rent adjustment. A multi-event design spanning several Canadian wildfires and multiple origin cities would separate the general receiving-market mechanism from the specific regional dynamics of British Columbia in 2017. That is the design this literature needs next.

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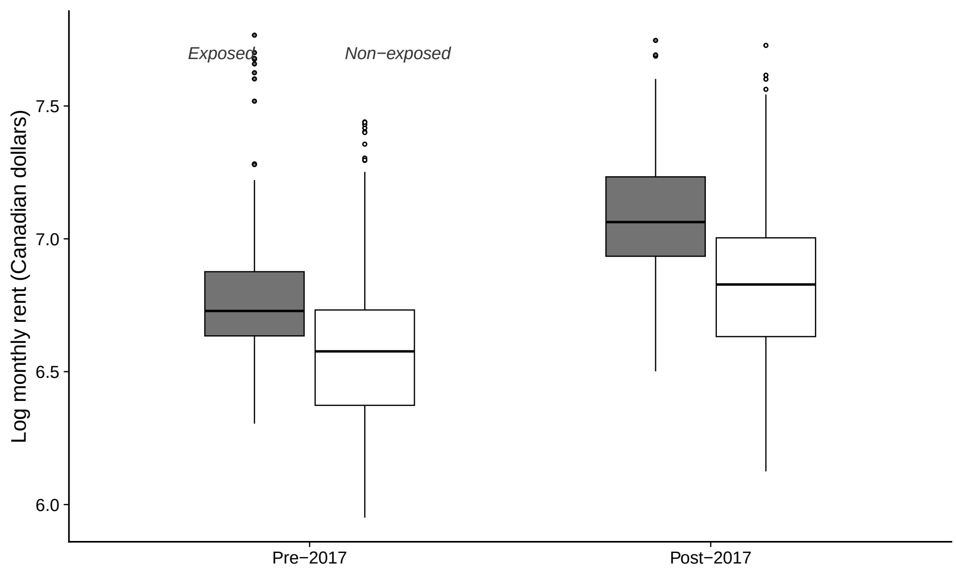
Appendix

A.1 Descriptive and Migration Evidence

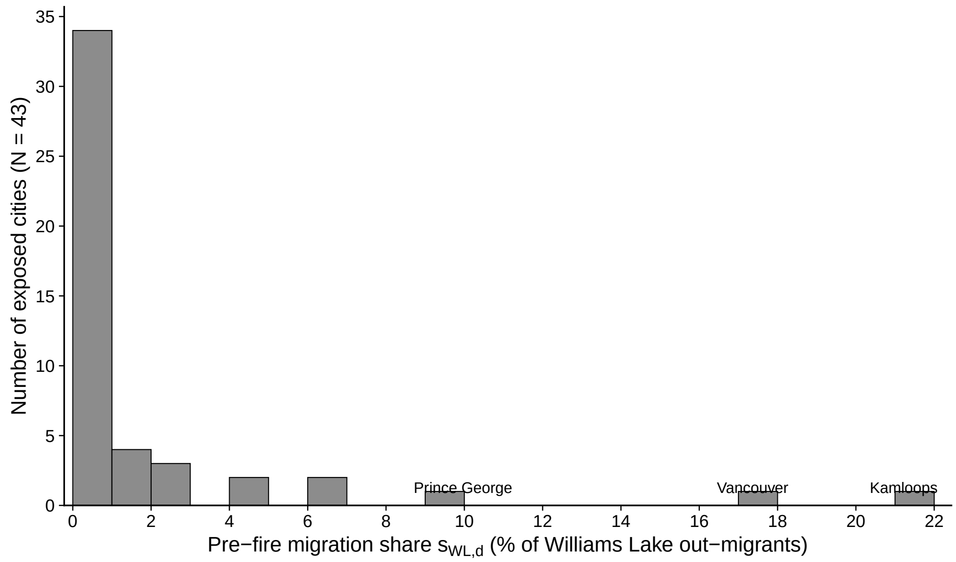
Appendix Figure 6 reports the distribution of log monthly rents by exposure status and period. Appendix Figure 7 shows the distribution of positive pre-fire Williams Lake migration shares across exposed cities. These figures document the raw structure of the outcome and the concentration of the exposure profile, but they do not carry the identification argument and are therefore reported outside the main text.

Figure 6: Distribution of Log Monthly Rent by Exposure Status and Period



Notes: Boxplots of log monthly rent over all city-year observations in the main sample, partitioned by exposure status ($s_{WL,d} > 0$ versus $s_{WL,d} = 0$) and by period (pre-2017: 2008–2016; post-2017: 2017–2024). The dependent variable is log monthly rent for two-bedroom apartments in buildings with at least three units.

Figure 7: Distribution of Pre-Fire Williams Lake Migration Shares

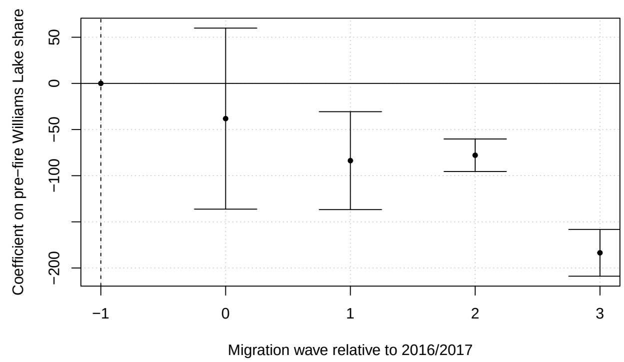


Notes: Histogram of pre-fire migration shares $s_{WL,d}$ across the 43 cities with strictly positive exposure. The share is defined as the fraction of Williams Lake out-migrants directed to city d during the 2016/2017 Statistics Canada migration wave, after excluding rural residual destinations and renormalising shares across retained city destinations. Cities with $s_{WL,d} = 0$ are not displayed.

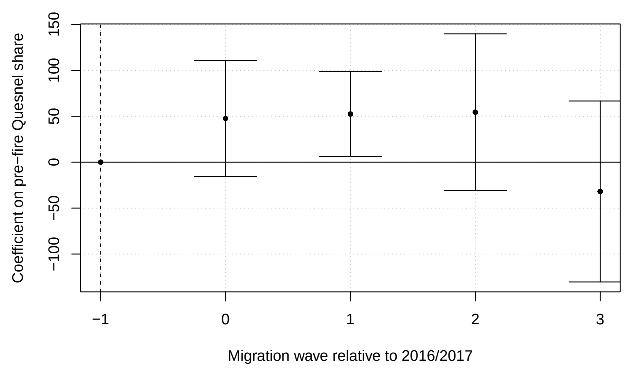
Appendix Figure 8repeats the same diagnostic for the Quesnel placebo origin. They visualise the flow evidence summarised in Section 4 without adding identification beyond Table 2.

Figure 8: Annual Arrivals Event Studies

Panel A. Williams Lake Origin



Panel B. Quesnel Placebo Origin



Notes: Panel A reports coefficients on $s_{WL,d}$ interacted with annual migration-wave indicators, from an OLS regression of annual bilateral arrivals from Williams Lake on destination-city and wave fixed effects. Panel B repeats the same diagnostic for Quesnel, a British Columbia forestry town that did not experience a wildfire in 2017. The omitted reference wave is 2016/2017. Shaded bands denote 95% confidence intervals. Standard errors are clustered at the destination-city level. Sample: waves 2016/2017–2020/2021.

A.2 Identification Diagnostics

Table 11 reports uniformly valid 95% confidence sets for the average post-2017 event-study effect under the honest-inference framework of Rambachan and Roth (2023). The summary statistic averages post-fire coefficients over $k = 1, \dots, 7$ and assigns zero weight to $k = 0$, because the 2017 rent observation is measured only a few months after the July evacuation.

The restriction parameter $\bar{M}$ governs how large post-treatment violations of parallel trends may be relative to the largest pre-treatment deviation. The case $\bar{M} = 0$ reproduces exact parallel trends. Larger values allow increasingly severe post-treatment departures from that benchmark. The breakdown value $\bar{M}^*$ is the smallest value at which the confidence set includes zero. In this application, $\bar{M}^* = 0.38$, which indicates that the positive post-2017 event-study pattern is not robust to even moderately larger violations than those observed before treatment.

Table 11: Sensitivity to Parallel-Trend Violations: Rambachan–Roth Confidence Sets

Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 *units.*
Rambachan–Roth (2023) sensitivity analysis.

Trend violation $\bar{M}$	Point estimate	95% confidence set
0.0	0.756	[0.35, 1.17]
0.5	—	[−0.16, 1.93]
1.0	—	[−0.94, 2.80]
1.5	—	[−1.83, 3.72]

Notes: Sensitivity analysis from Rambachan and Roth (2023) applied to the fully interacted rent event study with city and year fixed effects (N = 2,225). The target parameter is the average post-treatment effect over event times $k = 1, \ldots, 7$; $k = 0$ receives zero weight because the 2017 Canada Mortgage and Housing Corporation rent observation is measured in October, several months after the July evacuation. $\bar{M} = 0$ reports the conventional 95% confidence interval under exact parallel trends. For $\bar{M} > 0$, the table reports uniformly valid 95% confidence sets allowing post-treatment violations up to $\bar{M}$ times the largest pre-treatment deviation. The breakdown value is $\bar{M}^* = 0.38$, the smallest value at which the confidence set first includes zero. Standard errors clustered at the city level.

The main analysis remains in reduced form because the rental panel and the migration-flow panel do not share a comparable time structure. Canada Mortgage and Housing Corporation rents are observed annually from 2008 to 2024, while bilateral migration flows are available only for five waves, from 2016/2017 to 2020/2021. This appendix reports a limited-window fixed-effects 2SLS exercise to clarify why this paper does not convert the exposure estimates into an IV elasticity.

The exercise instruments observed annual arrivals from Williams Lake with $s_{WL,d} \times \mathbf{1}[\text{post-flow wave}]$ and then relates log rent to fitted arrivals. Two temporal alignments are shown. The concurrent alignment maps the 2017/2018 migration wave to rent year 2017. The lagged alignment maps that wave to rent year 2018. The lagged version better respects the fact that the migration wave extends through June, whereas Canada Mortgage and Housing Corporation rent is measured in October. Neither mapping is exact.

Table 13 reports the first stage, the reduced form over the same short window, and the resulting fixed-effects 2SLS coefficient. The first stage is precisely estimated but negative ($\hat{\eta} = -95.8^{***}$, $F = 85.9$). This reproduces the pattern documented in Section 4. High-share destinations record fewer annual arrivals from Williams Lake after 2017 than in the share-defining 2016/2017 wave. The short-window reduced form on rent is positive, with coefficients near 0.49 and 0.50 depending on the alignment. Dividing a positive reduced form by a negative first stage produces a negative fixed-effects 2SLS coefficient of approximately -0.005^{***}.

Table 12: Exploratory Limited-Window Migration First Stage and FE-2SLS

	Wave 2017/18 → Rent 2017			Wave 2017/18 → Rent 2018		
	(1) First stage	(2) Reduced form	(3) FE-2SLS	(4) First stage	(5) Reduced form	(6) FE-2SLS
Dependent variable:	*Arrivals*	*Log rent*	*Log rent*	*Arrivals*	*Log rent*	*Log rent*
$s_{WL,d}$ × PostFlow	-95.80^{***}	0.492^{***}		-95.80^{***}	0.497^{***}	
	(8.940)	(0.134)		(8.940)	(0.137)	
Arrivals from Williams Lake			-0.005^{***}			-0.005^{***}
			(0.002)			(0.002)
Observations	655	655	655	655	655	655
R^2	0.97	0.98	0.98	0.97	0.97	0.97
Within R^2	0.12	0.02	0.02	0.12	0.02	0.02
RMSE	2.838	0.041	0.043	2.838	0.043	0.046
First-stage F			85.9			85.9
Destination FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓

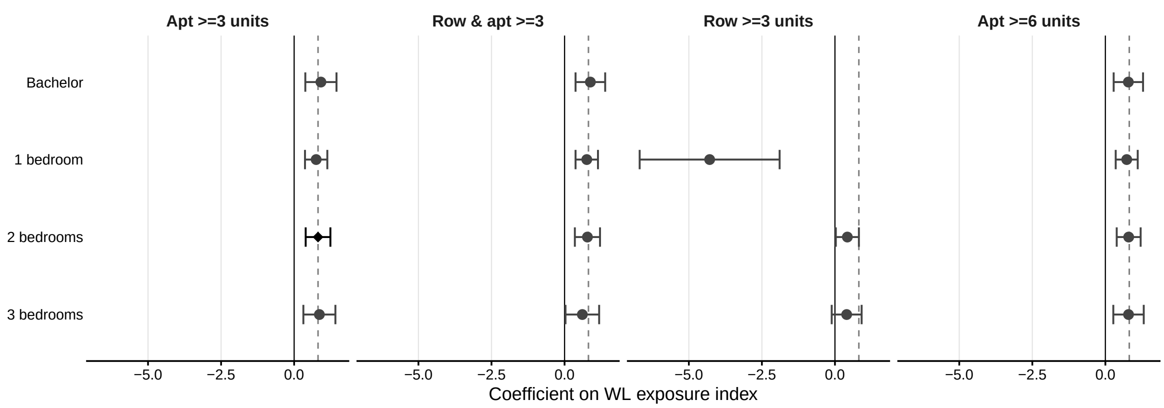
Notes: Columns (1) and (4) report first-stage regressions where the dependent variable is annual migration arrivals from Williams Lake. Columns (2) and (5) report reduced-form regressions on log monthly rent for two-bedroom apartments in buildings ≥ 3 units. Columns (3) and (6) report fixed-effects 2SLS estimates, instrumenting observed Williams Lake arrivals with $s_{WL,d}$ × PostFlow. Columns (1)–(3) align migration wave 2017/2018 with rent year 2017; columns (4)–(6) align the same wave with rent year 2018. All specifications include destination-city and year fixed effects. Standard errors clustered at the destination-city level in parentheses. These estimates are exploratory and are not used as the preferred rent-effect specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

That coefficient has no credible structural interpretation in this setting. It does not imply that additional Williams Lake arrivals lower rents in receiving cities. Instead, it shows that the available annual migration-flow measure is not a suitable endogenous regressor for an IV design built around a short-lived emergency evacuation. The negative first stage also signals a monotonicity failure. Cities with stronger pre-fire ties to Williams Lake attracted fewer evacuees post-fire relative to the share-defining wave, leaving no complier population and no coherent local average treatment effect to recover. The limited flow window also inherits the regional confounding concerns that affect the main reduced-form design. For these reasons, the fixed-effects 2SLS exercise is reported as a diagnostic that limits interpretation, not as a preferred estimate of the effect of displaced households on rents.

A.3 Rental-Market Heterogeneity

Figure 9 reports the complete heterogeneity results discussed briefly in Section 5. Each point is the coefficient on Z_{WL} from a separate regression with city and year fixed effects. Vertical lines show 95% confidence intervals. Standard errors clustered at the city level. The dashed reference line marks the baseline estimate of 0.818 (2 bedrooms, apartment structures ≥ 3 units). The row structure column omits the bachelor-unit cell, which is not estimable after balancing ($N_d = 2$); the 1-bedroom estimate in that column (-4.288, s.e. 1.222) is driven by only 14 cities and is not interpretable as an economic estimate.

Figure 9: Heterogeneity by Unit Type and Housing Structure



Notes: Each point is the OLS coefficient on the post-2017 Williams Lake exposure index Z_{WL} from a separate regression of log monthly rent on Z_{WL} with city and year fixed effects. Bars show 95% confidence intervals. Standard errors clustered at the city level. The dashed vertical line marks the baseline estimate (0.818, 2 bedrooms, apt ≥ 3 units, diamond). Period: 2008–2024.

The supply-elasticity estimates in Saiz (2010) imply that rent responses to housing-demand shocks are two to three times larger in cities where supply is physically or regulatorily constrained. This prediction is directly testable in the shift-share design by interacting the exposure index with a proxy for supply inelasticity. Because the Canadian equivalent of the Wharton Residential Land Use Regulatory Index is not available for all 131 cities,

Table 13: Exploratory Limited-Window Migration First Stage and FE-2SLS

	Wave 2017/18 → Rent 2017			Wave 2017/18 → Rent 2018		
	(1) First stage	(2) Reduced form	(3) FE-2SLS	(4) First stage	(5) Reduced form	(6) FE-2SLS
Dependent variable:	*Arrivals*	*Log rent*	*Log rent*	*Arrivals*	*Log rent*	*Log rent*
$s_{WL,d}$ × PostFlow	−95.80***	0.492***		−95.80***	0.497***	
	(8.940)	(0.134)		(8.940)	(0.137)	
Arrivals from Williams Lake			−0.005***			−0.005***
			(0.002)			(0.002)
Observations	655	655	655	655	655	655
R^2	0.97	0.98	0.98	0.97	0.97	0.97
Within R^2	0.12	0.02	0.02	0.12	0.02	0.02
RMSE	2.838	0.041	0.043	2.838	0.043	0.046
First-stage F			85.9			85.9
Destination FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓

Notes: Columns (1) and (4): first-stage regressions where the dependent variable is annual migration arrivals from Williams Lake. Columns (2) and (5): reduced-form regressions on log monthly rent for two-bedroom apartments in buildings ≥ 3 units. Columns (3) and (6): FE-2SLS second stage; observed Williams Lake arrivals instrumented by $s_{WL,d}$ × PostFlow. Columns (1)–(3) align migration wave 2017/2018 with rent year 2017; columns (4)–(6) align the same wave with rent year 2018. All specifications include destination-city and year fixed effects. Standard errors clustered at the destination-city level in parentheses. These estimates are exploratory. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I use the average pre-fire rent level (2008–2016) as a proxy. Cities with historically higher rents face more supply-constrained conditions, consistent with a Saiz-type mechanism.

Let $\text{Tight}_d = \mathbf{1}[\bar{R}_{d,\text{pre}} \geq \$775]$, where \$775 is the sample median pre-fire rent. The extended specification is:

$$\log R_{dt} = \alpha + \pi_1 Z_{WL,dt} + \pi_2 \left(Z_{WL,dt} \times \text{Tight}_d\right) + \gamma_d + \lambda_t + \varepsilon_{dt}. \tag{5}$$

Equation (5) nests the baseline specification (3) when $\pi_2 = 0$. The Saiz prediction is $\hat{\pi}_2 > 0$.

Table 14 reports the results. The interaction term is $\hat{\pi}_2 = 0.007$ (s.e. 0.547), statistically indistinguishable from zero. The point estimates across tight and slack subsamples are nearly identical (0.836 vs. 0.784); the difference in statistical significance between the two subsamples reflects sample size and precision, not heterogeneous treatment effects. The Saiz (2010) supply-elasticity prediction is therefore not validated by this design.

This null is itself informative. Two interpretations are consistent with it. First, the residual identifying variation, once city and year fixed effects are absorbed, may be insufficient to detect effect heterogeneity of plausible magnitude across the two subsamples. Second,

Table 14: Supply-Side Heterogeneity: Tight vs. Slack Markets

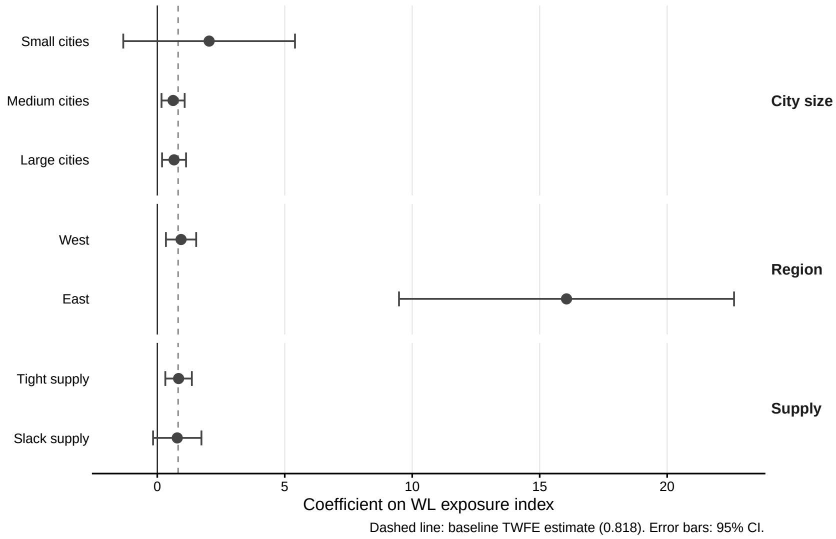
	(1) Interaction	(2) Tight markets	(3) Slack markets
Z_{WL}	0.812	0.836***	0.784
	(0.504)	(0.265)	(0.484)
$Z_{WL} \times \text{Tight}_d$	0.007		
	(0.547)		
Observations	2,225	1,122	1,103
R^2	0.940	0.865	0.944
Within R^2	0.021	0.026	0.010
City FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: This table reports reduced-form exposure estimates. The dependent variable is log monthly rent for two-bedroom apartments in buildings with at least three units. Tight markets are cities with mean 2008–2016 rent above the sample median ($775). Column (1) estimates the pooled interaction specification with $Z_{WL,dt} \times \text{Tight}_d$, where $\text{Tight}_d = \mathbf{1}[\bar{R}_{d,\text{pre}} \geq \$775]$. Columns (2)–(3) estimate the baseline specification separately for tight and slack markets. All specifications include city and year fixed effects. Standard errors are clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

if the baseline estimate reflects provincial confounding rather than a genuine displacement channel, there is no theoretical reason to expect it to be larger in supply-constrained markets; regional rent trends need not respect supply-elasticity gradients. The null interaction result is therefore consistent with the regional confounding hypothesis as well as with a genuine but homogeneous displacement effect.

Figure 10 plots the exposure coefficient across three remaining heterogeneity dimensions, covering city size (small, medium, large), geographic region (West, East), and housing supply tightness following Saiz (2010). The figure reports all subgroup cuts regardless of statistical significance.

Figure 10: Heterogeneity by City Size, Region, and Housing Supply Constraints



Notes: Each point reports the reduced-form coefficient on the post-2017 Williams Lake exposure index across seven subgroups. Bars show 95% confidence intervals. Standard errors are clustered at the city level. The dashed vertical line marks the baseline reduced-form estimate (0.818, full sample, column 4 of Table 3). City size is defined as small (<50,000), medium (50,000–200,000), and large (>200,000). Region is West (BC, AB, SK, MB) or East. Supply tightness is defined by pre-2017 average rent above or below the sample median, following Saiz (2010). Benjamini-Hochberg adjusted p-values are reported in the figure source file; five of seven subgroup estimates survive correction at $q < 0.05$. Period: 2008–2024.

A.4 Exposure, Influence, and Robustness

Table 15 reports Rotemberg weights for each exposed city. In this single-shock shift-share design, the weight of city d is proportional to $s^2_{WL,d}$ (Goldsmith-Pinkham et al., 2020). Kamloops and Vancouver together account for roughly 40 percent of total identification weight; the top five cities account for approximately 60 percent.

The economic implication of this concentration is stark. The coefficient $\hat{\pi}$ is effectively a weighted average of post-2017 rent differentials across exposed cities, with Kamloops and Vancouver together accounting for nearly half the signal. Kamloops is a mid-sized Interior city that also experienced strong population growth and housing demand from in-migration unrelated to the wildfire. Vancouver is Canada's most supply-constrained rental market and was simultaneously affected by the Foreign Buyers Tax (August 2016) and subsequent demand displacement toward rentals. If either city's post-2017 rent trajectory reflects forces

Table 15: Rotemberg Weights by Pre-Fire Exposure Rank

City	$s_{WL,d}$ (%)	$s^2_{WL,d}$	w^R_d (%)	Cumulative (%)
Kamloops	21.22	0.0450	43.97	43.97
Vancouver	17.36	0.0301	29.43	73.40
Prince George	9.97	0.0099	9.70	83.10
Kelowna	6.91	0.0048	4.67	87.77
Quesnel	6.75	0.0046	4.45	92.22
Abbotsford–Mission	4.98	0.0025	2.43	94.65
Vernon	4.18	0.0017	1.71	96.35
Calgary	2.57	0.0007	0.65	97.00
Victoria	2.57	0.0007	0.65	97.64
Campbell River	2.41	0.0006	0.57	98.21
All other cities ($N = 38$)	21.06	—	1.79	100.00

Notes: Rotemberg weights $w^R_d = s^2_{WL,d} / \sum_{d'} s^2_{WL,d'}$ (Goldsmith-Pinkham et al., 2020). In this balanced panel with a single binary Post indicator, these weights equal the squared pre-fire migration share normalised to sum to one across the 43 exposed cities. Each weight measures city d's contribution to the overall identification of $\hat{\pi}$. Non-exposed cities ($s_{WL,d} = 0$) carry zero Rotemberg weight and serve as the comparison group, not as identifying variation.

specific to that city rather than the wildfire exposure channel, the aggregate coefficient will be contaminated. The leave-one-out diagnostics in Figure 12 show that removing Kamloops raises rather than lowers the estimate, which argues against a single-city story but does not resolve the broader identification question. The honest interpretation is that $\hat{\pi}$ is largely identified by what happened in two or three cities with particular economic histories during a period of broader provincial rental-market pressure.

Figure 12 reports the leave-top-k influence diagnostics. The coefficient rises monotonically from the baseline (0.818) as the most exposed cities are dropped. Removing Kamloops alone raises it to 1.100, and dropping the ten most exposed cities yields 5.420 (s.e. 2.700). This pattern reflects a non-monotone relationship between exposure and the post-2017 rent outcome. Once Kamloops is excluded, medium-exposure cities in the range $s_{WL,d} \in [0.03, 0.10]$ dominate and exhibit a stronger per-unit relationship. As the top-k cities are progressively removed, the exposure variance in the denominator shrinks mechanically, inflating the normalised coefficient. The result argues against a single-city story but does not resolve the regional confounding concern.

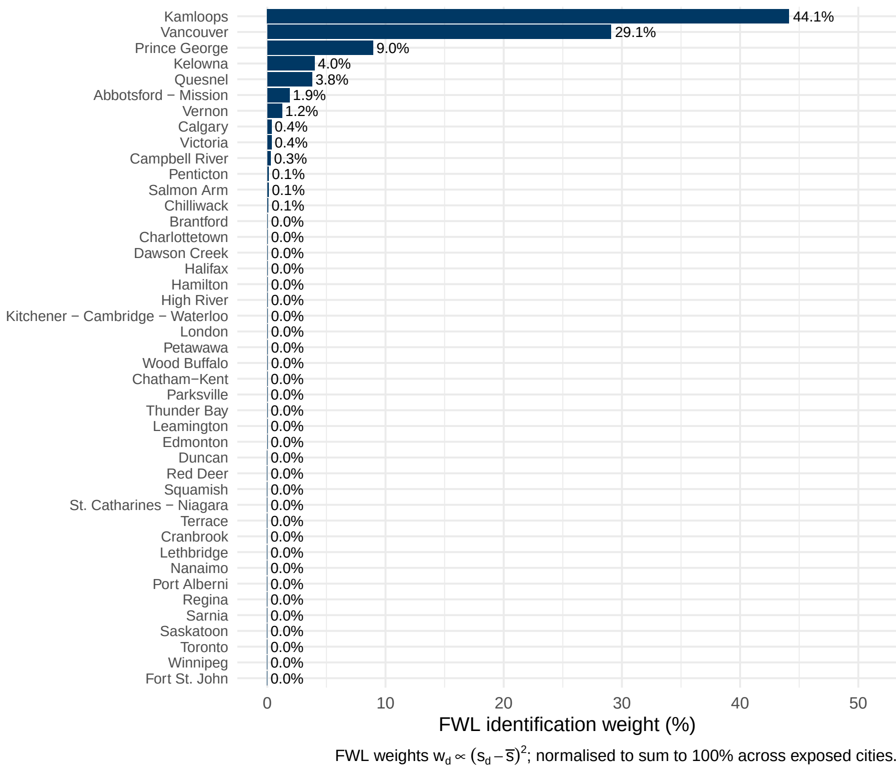
A.4a TWFE Weights and the Negative-Weight Concern

de Chaisemartin and D'Haultfœuille (2020) show that the two-way fixed effects estimator can assign negative weights to some average treatment effects when those effects are heterogeneous across treatment intensities. For the continuous exposure index $Z_{WL,dt} = s_{WL,d} \times \mathbf{1}[t \geq 2017]$, this concern is addressed analytically. By the Frisch-Waugh-Lovell theorem, the TWFE estimator projects $\log R_{dt}$ onto the within-transformed regressor $\tilde{Z}_{WL,dt} = Z_{WL,dt} - \bar{Z}_{WL,d} - \bar{Z}_{WL,t} + \bar{Z}_{WL}$. Substituting $Z_{WL,dt} = s_{WL,d} \times \text{Post}_t$ yields

$$\tilde{Z}_{WL,dt} = (s_{WL,d} - \bar{s}) \times \left(\text{Post}_t - \frac{T^{\text{post}}}{T}\right),$$

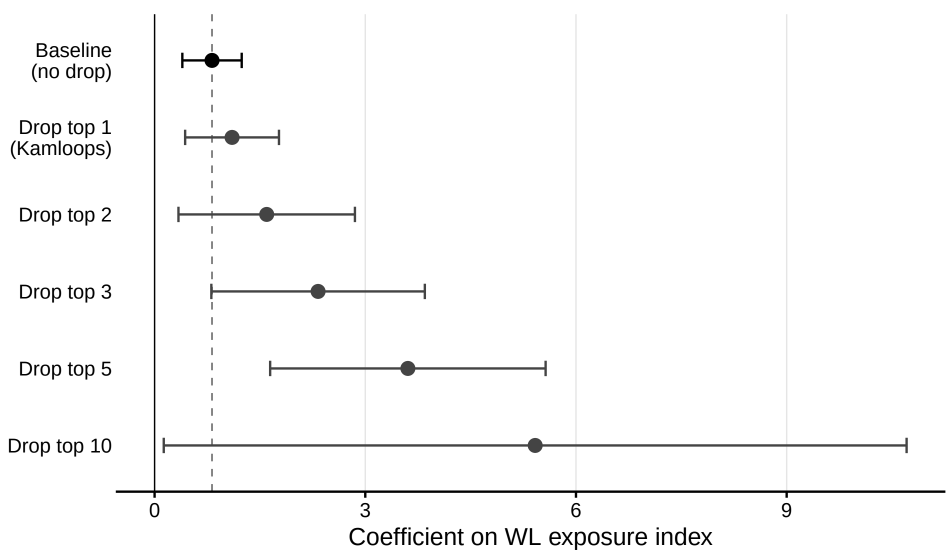
where $\bar{s}$ is the cross-city mean of the exposure share and $T^{\text{post}}/T = 8/17$ for the 2008–2024 panel. The TWFE weight of cell (d, t) is $w_{dt} \propto \tilde{Z}^2_{WL,dt} \geq 0$ for all (d, t). The simultaneous shock and the time-invariant post-treatment intensity guarantee non-negative weights by construction, so $\hat{\pi}$ is a non-negatively weighted average of city-level rent differentials. Figure 11 displays the implied weight distribution across exposed cities, which closely mirrors the Rotemberg weights reported in Table 15: Kamloops and Vancouver together account for approximately 77 percent of the identifying variation. The replication script (`twfe_diag.R`) verifies this result numerically.

Figure 11: TWFE Identification Weights by Exposed City



Notes: Frisch-Waugh-Lovell identification weights $w_d \propto (s_{WL,d} - \bar{s})^2$, normalised to sum to 100 percent across the 43 exposed cities ($s_{WL,d} > 0$). Non-exposed cities carry zero weight and serve as the comparison group. $\bar{s} = 0.0074$ (cross-city mean of the pre-fire exposure share). Period: 2008–2024.

Figure 12: Leave-One-Out Influence Diagnostics



Notes: Each point is the OLS coefficient on Z_{WL} after sequentially dropping the top k cities by pre-fire exposure rank, with 95% confidence intervals. The dashed vertical line marks the baseline estimate (0.818, full sample). Standard errors clustered at the city level. Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. Period: 2008–2024.

Table 16 collects secondary exposure and sample checks, including the Fort McMurray-only estimate, the joint WL+FM specification, the aggregated exposure index, and the unbalanced-panel re-estimation.

Table 16: Robustness: Alternative Exposure Specifications

Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. Reduced-form exposure specification estimated by OLS..

	(1) Baseline	(2) FM only	(3) WL+FM	(4) Aggregated	(5) Unbalanced
Z_{WL} (BC 2017)	0.818***		0.835***		0.828***
	(0.216)		(0.234)		(0.218)
Z_{FM} (Fort McMurray 2016)		0.055	−0.670		
		(1.000)	(0.813)		
$\hat{Z}$ (aggregated)				0.637***	
				(0.169)	
Observations	2,225	2,242	2,242	2,242	2,382
R^2	0.94	0.95	0.95	0.95	0.94
Within R^2	0.02	<0.001	0.03	0.02	0.02
RMSE	0.075	0.069	0.068	0.069	0.075
City FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓

Notes: Dependent variable: log monthly rent, two-bedroom apartments, buildings ≥ 3 units. OLS: ordinary least squares. FE: fixed effects. FM: Fort McMurray. WL: Williams Lake. $Z_{WL} = s_{WL,d} \times \text{Post}_{2017}$: Williams Lake shift-share exposure index (observed pre-fire shares). Z_{FM}: Fort McMurray exposure index (gravity-predicted shares $\times$ Post_{2016}). $\hat{Z} = Z_{WL} + Z_{FM}$: aggregated exposure. Column (1): main panel of 131 cities (N = 2,225). Columns (2)–(4): panel augmented with Fort McMurray destinations (N = 2,242). Column (5): unbalanced panel (N = 2,382). All specifications include city and year fixed effects. Standard errors clustered at the city level in parentheses. Period: 2008–2024.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17 verifies that the baseline inference is not sensitive to the choice of clustering level. The baseline clusters at the city level (131 clusters). Province-level clustering (10 clusters) and two-way clustering at the city-by-year level are both more demanding and produce slightly wider standard errors, but the coefficient on Z_{WL} remains statistically significant at the one-percent level across all specifications. Population-weighted and rent-weighted OLS yield point estimates close to the baseline. The evidence in this table supports reading the main result as an inference that is stable across plausible clustering assumptions.

Table 17: Robustness: Alternative Standard Errors and Weighting

Dependent variable: log monthly rent, two-bedroom apartments, buildings $\geq$ 3 units. Reduced-form exposure specification estimated by OLS..

	(1) Baseline city cluster	(2) Cluster: province	(3) Cluster: city $\times$ year	(4) WLS population	(5) WLS rent 2008
Z_{WL} (BC 2017)	0.818***	0.818***	0.818***	0.543***	0.892***
	(0.216)	(0.157)	(0.235)	(0.084)	(0.248)
Observations	2,225	2,225	2,225	2,225	2,225
R^2	0.940	0.940	0.940	0.977	0.917
Within R^2	0.021	0.021	0.021	0.075	0.021
City FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓

Notes: Dependent variable: log monthly rent, two-bedroom apartments, buildings $\geq$ 3 units. Column (1): baseline specification with standard errors clustered at the city level (131 clusters). Column (2): standard errors clustered at the province level (10 clusters). Column (3): two-way clustering at city and year levels. Columns (4)–(5): weighted least squares; column (4) weights by city population, column (5) by 2008 rent level. Point estimates are identical across columns (1)–(3) because the coefficient is the same regardless of the variance estimator; the spread in standard errors reflects clustering assumptions only. All specifications include city and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

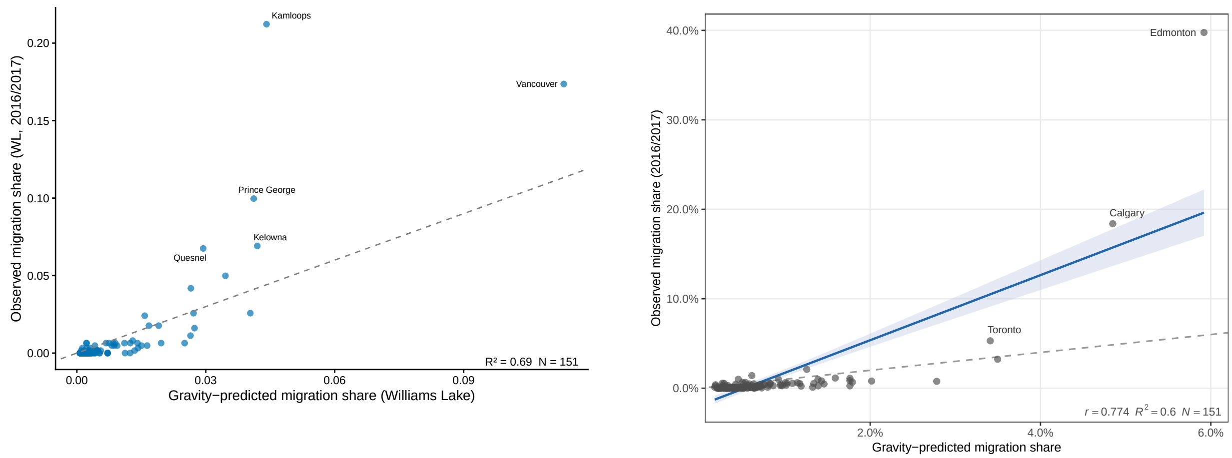
A.5 Fort McMurray Contrast

The Fort McMurray exercise remains a secondary comparison. It documents a contrast between Williams Lake exposure and a gravity-predicted Fort McMurray exposure profile, but it does not identify a duration mechanism. The gravity-predicted Fort McMurray exposure profile is used as a descriptive comparison; it is not interpreted as establishing a causal first stage.

Appendix Figure 13 compares observed and gravity-predicted migration shares for Williams Lake and Fort McMurray. The comparison is diagnostic: it shows whether the exposure profiles are consistent with standard gravity forces rather than idiosyncratic destination choices.

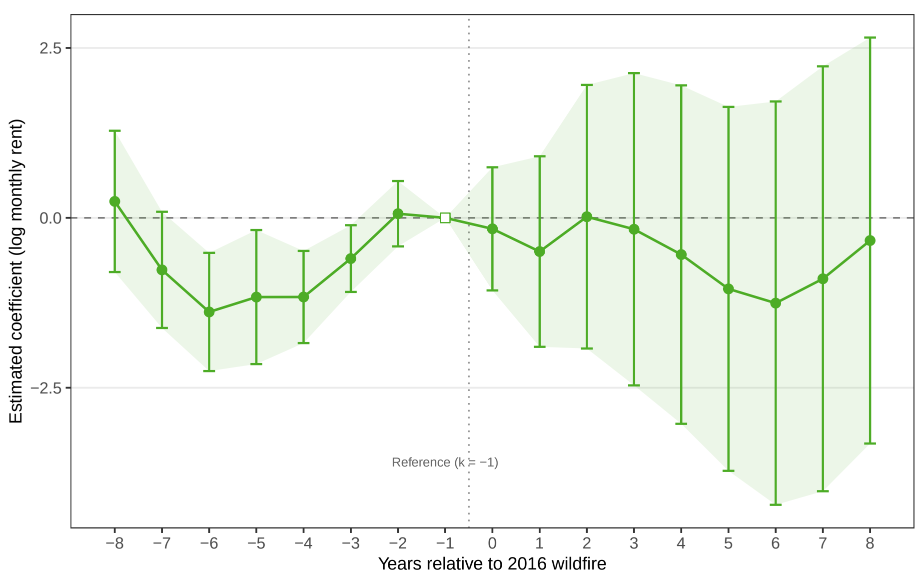
Figure 13: Gravity Model Validation for Wildfire-Origin Migration Shares

Panel A. Williams Lake **Panel B. Fort McMurray**



Notes: Panel A plots observed pre-fire Williams Lake migration shares against gravity-predicted shares constructed from destination population and bilateral distance. Panel B repeats the same exercise for Wood Buffalo (Fort McMurray). Each point is a city in the analytical sample. Gravity predictions are used as exposure diagnostics and are not interpreted as causal first stages.

Figure 14: Event Study: Fort McMurray 2016



Notes: Coefficients from a fully interacted specification regressing log monthly rent on gravity-predicted Fort McMurray exposure interacted with year indicators, with city and year fixed effects. The omitted reference year is 2015. Shaded bands denote 95% confidence intervals. Standard errors clustered at the city level. Dependent variable: log monthly rent for two-bedroom apartments in buildings with at least three units. Period: 2008–2024.